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CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

OptiPack AI – AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization (FA26SE036)

– Ho Chi Minh City, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
24/10/2026	A	ThuanNPM	Initial SRS for Review 2: product overview, actors, use cases, system functional overview, functional requirements of 12 features, non-functional requirements, business rules and message list.
25/10/2026	M	ThuanNPM	Aligned with the code: two-factor authentication is optional for every role (not mandatory for Admin); order sync runs per shop interval (default 5 minutes), not every minute; manual sync is Admin only; Tiki / TikTok adapters are not registered; Dashboard Overview partly uses sample data; AI text service is Groq; packing-plan performance measured by benchmark script; mobile app noted as planned for Iteration 3.

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

Small and medium online sellers in Vietnam sell the same products on several channels at once. Orders arrive separately from each marketplace, staff pick items from memory, pack them in whatever box is at hand and book carriers by hand. The result is mis-picked items, oversized boxes, wasted packaging material, unnecessary shipping cost and no single place to follow an order from purchase to delivery or return.

OptiPack AI is a web-based fulfillment system for one seller (one Store Owner and its warehouse team). It pulls orders from connected marketplaces into one queue, groups orders that go to the same recipient, reserves stock, guides warehouse staff to pick by bin and by oldest lot (FIFO), computes a 3D packing plan for every order (box choice, item placement, filling materials) with a step-by-step packing guide, books one carrier shipment per parcel with GHN, follows delivery, failed delivery and return to the warehouse, and handles customer return and exchange requests. Every important status change is written back to the marketplace so that end customers can track their orders.

The release covered by this SRS integrates two sales channels:

- **Lazada** – real marketplace connected through the Lazada Open Platform (OAuth; a scheduler runs every minute and synchronizes each shop when its configured interval has passed, default 5 minutes). OptiPack AI does not write order status back to Lazada (read-only by design).
- **AURELLE** – a fashion web store built by the team (customer website + Open API that follows the Lazada API style, HMAC-signed requests and webhooks). AURELLE plays the role of a second marketplace and lets the team demonstrate the full loop: order → fulfillment → customer tracking → return / exchange.
- **Tiki, TikTok Shop, Shopee and Facebook Marketplace** – out of scope of this release because of partner API access constraints (see Form BM03). Adapter source files for Tiki and TikTok Shop remain in the repository but are not registered, so these shops cannot be connected.

The context diagram below shows the boundary of OptiPack AI and the external entities it exchanges data with.

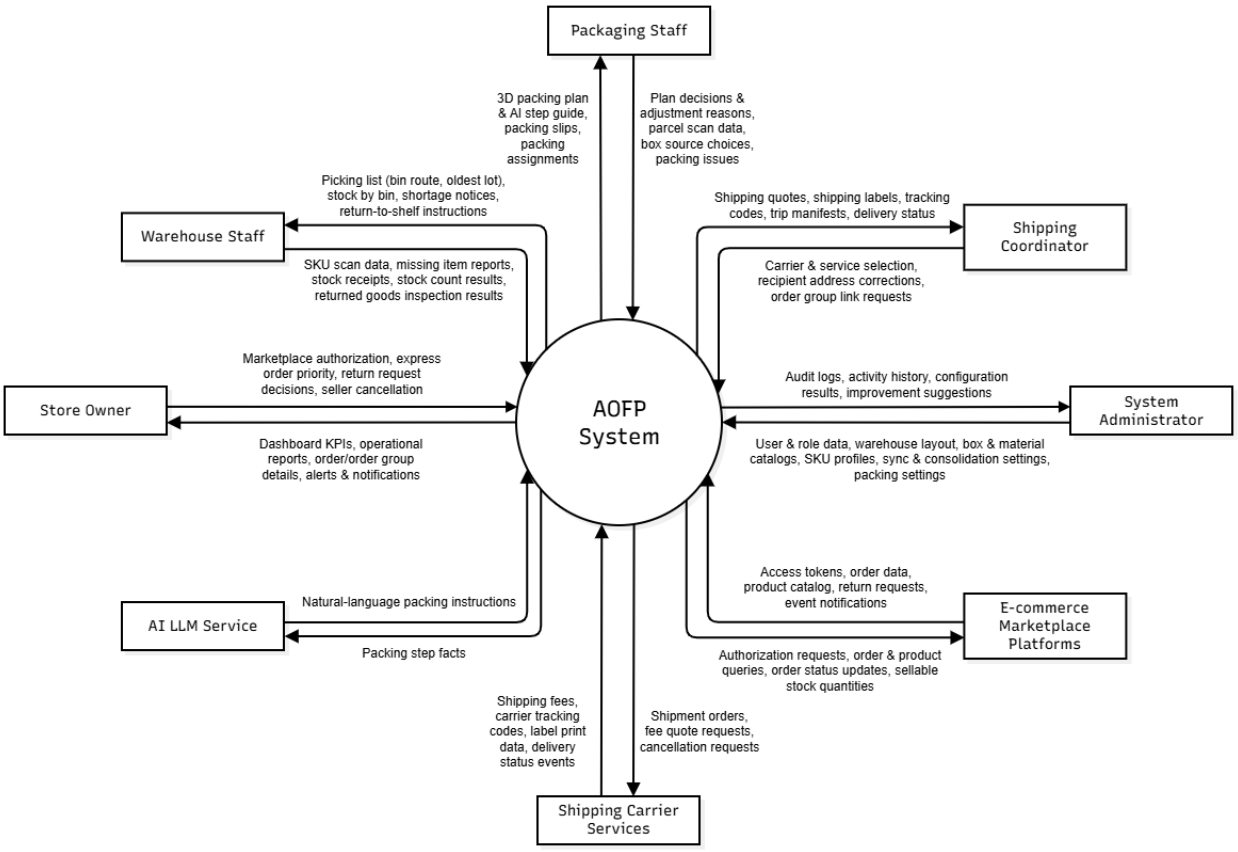


Figure 1. Context diagram of OptiPack AI

External entity	Type	Data exchanged with OptiPack AI
Lazada Open Platform	External system	OAuth tokens, orders and order items, product catalog (seller SKUs)
AURELLE Website / Open API	External system (team-built marketplace)	Orders, order-changed / product-updated / return-requested webhooks, return requests, order status and return status written back, available stock pushed every minute
GHN (Giao Hàng Nhanh)	External system (carrier)	Shipping fee quotes, shipping orders per parcel, tracking codes, status webhooks, A5 labels
Google Identity	External system	OAuth 2.0 login for staff whose e-mail already exists in the system
Gmail SMTP	External system	Account, temporary password, password reset and lock notification e-mails
AI text service (Groq API, OpenAI-compatible)	External system	Packing-guide wording for a parcel (no personal data)
Staff (5 roles)	People	Operate the web portal: configuration, picking, packing, shipping, returns
Customer (AURELLE)	People	Places orders, tracks delivery, requests return / exchange on the AURELLE website

2. User Requirements

2.1 Actors

#	Actor	Description
1	Admin	Full access. Manages user accounts and roles, connects / disconnects marketplace shops, configures warehouses, internal SKUs, packaging catalog, carriers and

#	Actor	Description
		system settings. Can enable optional two-factor authentication (TOTP), like every staff account.
2	Store Owner	Business owner. Monitors orders and order groups, sets express priority, cancels AURELLE orders as seller, approves or rejects return requests, configures packing and shipping settings, manages staff leave, views reports and action history.
3	Warehouse Staff	Picks items for assigned order groups, reports missing items, receives stock by lot, assigns SKUs to bins, adjusts / transfers / counts stock, receives returned parcels and inspects returned items.
4	Packaging Staff	Reviews, approves or rejects packing plans, packs parcels (scan, seal, weigh), reviews parcels with abnormal weight, decides partial shipment when items are missing.
5	Shipping Coordinator	Compares carrier quotes, creates shipments (single group or same-recipient trip), corrects delivery addresses, follows delivery and failed delivery, links or splits / merges order groups.
6	Customer	End buyer on the AURELLE website. Registers, browses products, places and cancels orders, tracks delivery and requests return or exchange.
7	AURELLE Web Admin	Administrator of the AURELLE website. Manages storefront categories, products, variants and images, and Open Platform developer apps (app key / secret).
8	Lazada Open Platform	External system providing orders and catalog of the connected Lazada shop.
9	AURELLE Open Platform	External system providing AURELLE orders, return requests and webhooks, and receiving status and stock written back by OptiPack AI.
10	GHN Carrier	External carrier system that quotes fees, creates shipping orders and reports delivery status.
11	System Scheduler	Time-based jobs inside OptiPack AI (order sync, packing-plan computation, status write-back, stock push, SLA checks).

2.2 Use Cases

2.2.1 Diagram(s)

Use cases are grouped into packages so that each diagram stays readable. One diagram is drawn per package below.

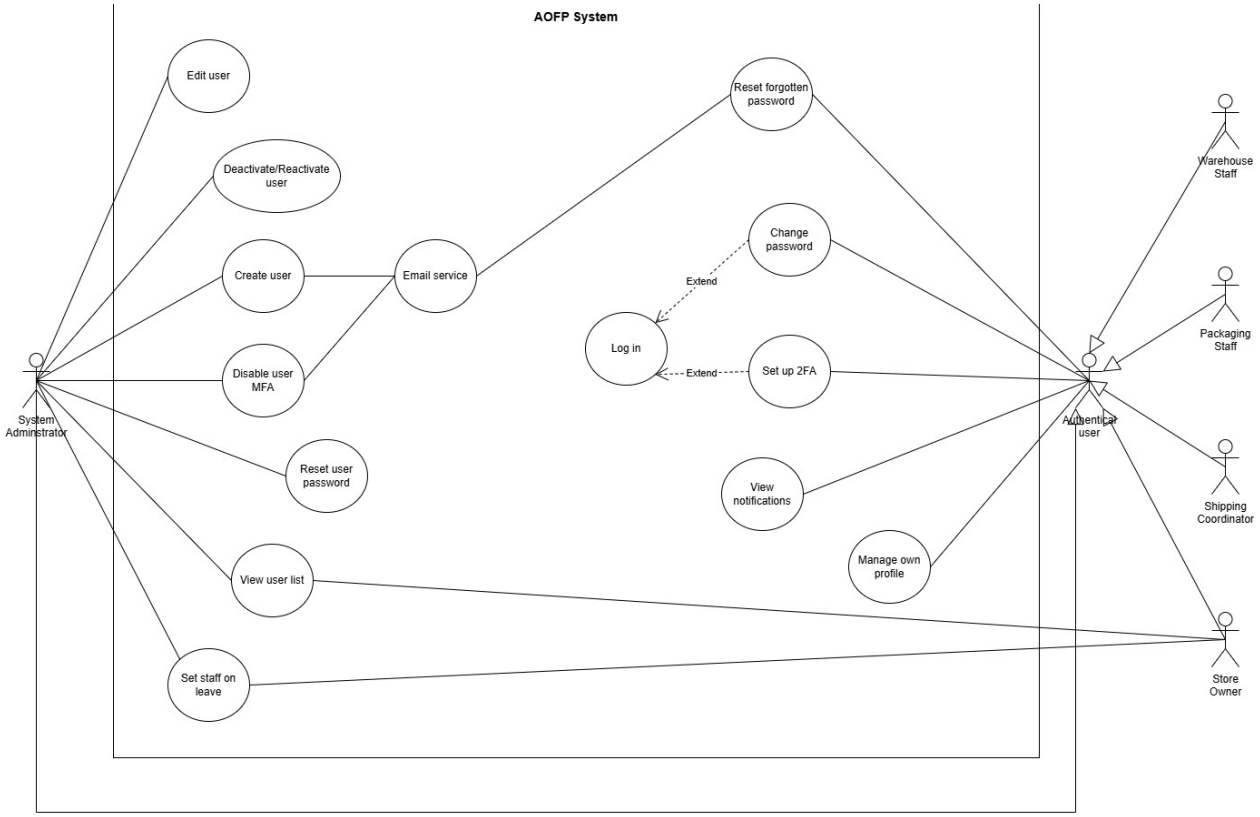


Figure 2. Use case diagram – Account & Security (UC-01 to UC-08)

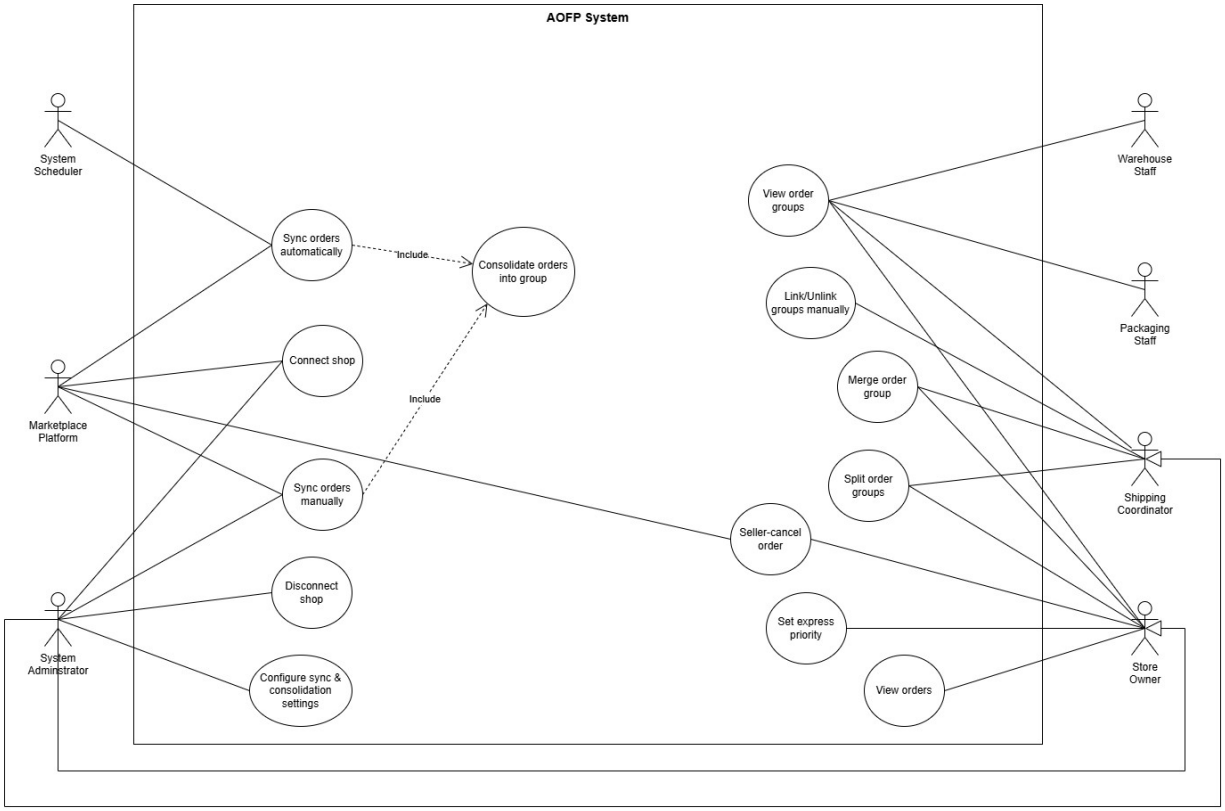


Figure 3. Use case diagram – Marketplace & Orders (UC-09 to UC-19)

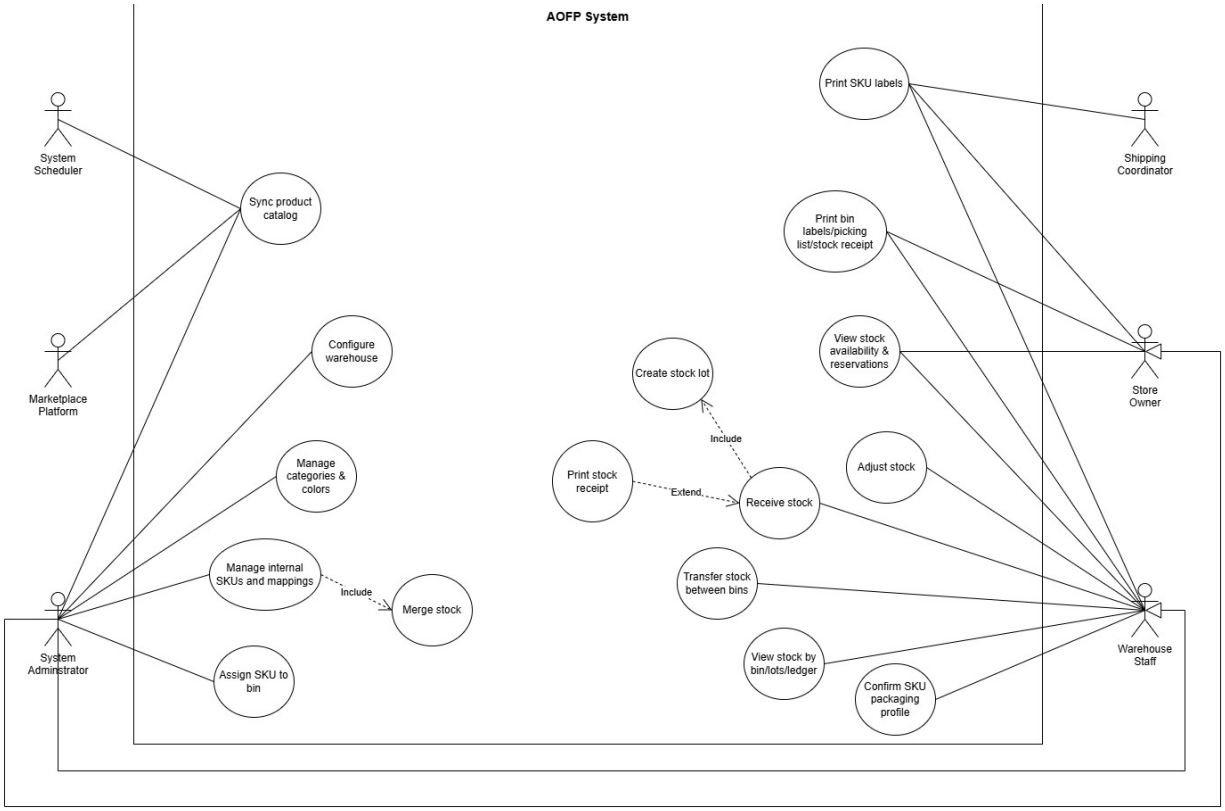


Figure 4. Use case diagram – Warehouse & Inventory (UC-20 to UC-25)

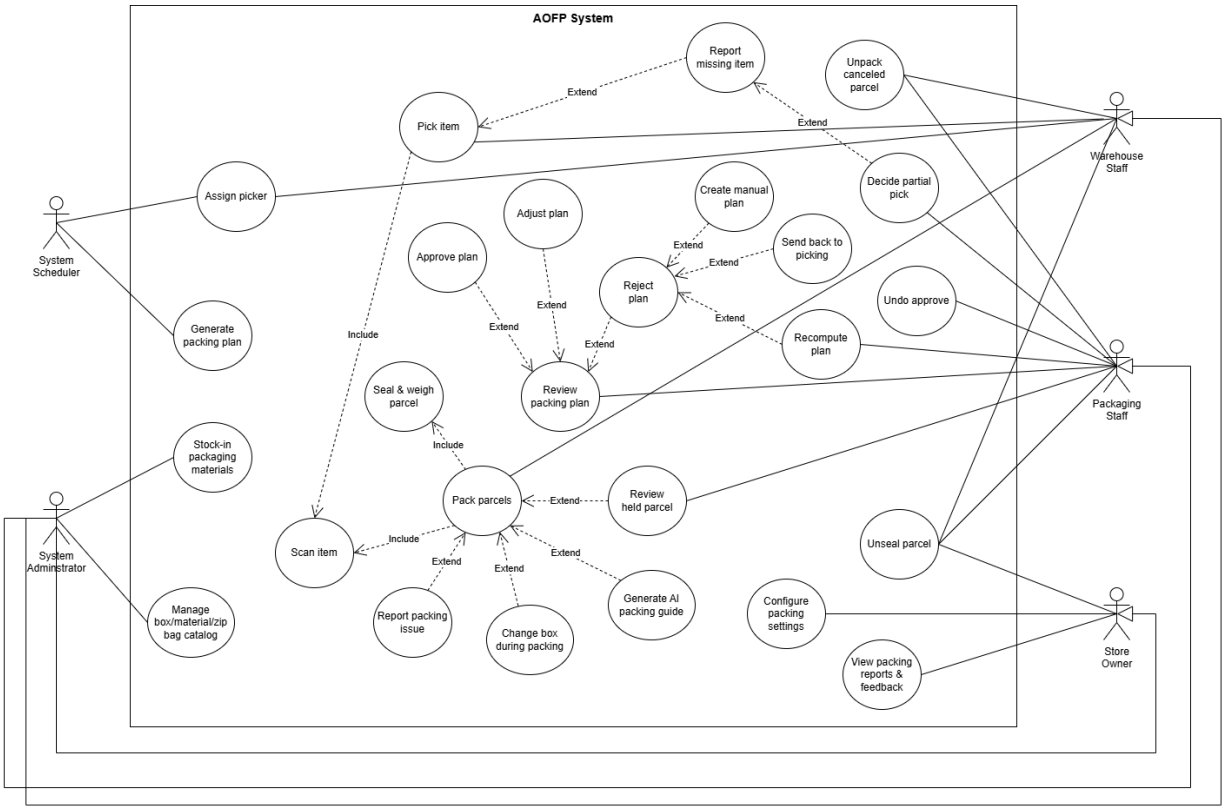


Figure 5. Use case diagram – Picking & Packing (UC-26 to UC-36)

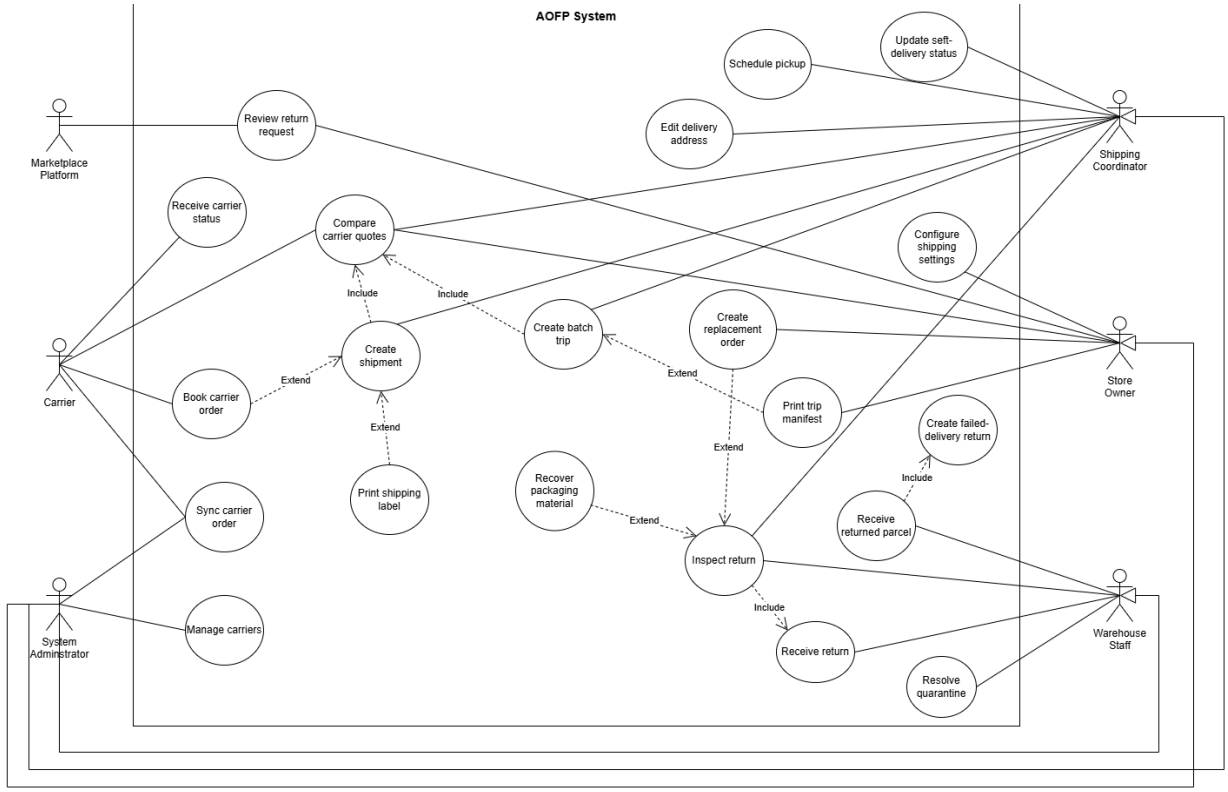


Figure 6. Use case diagram – Shipping & Returns (UC-37 to UC-44)

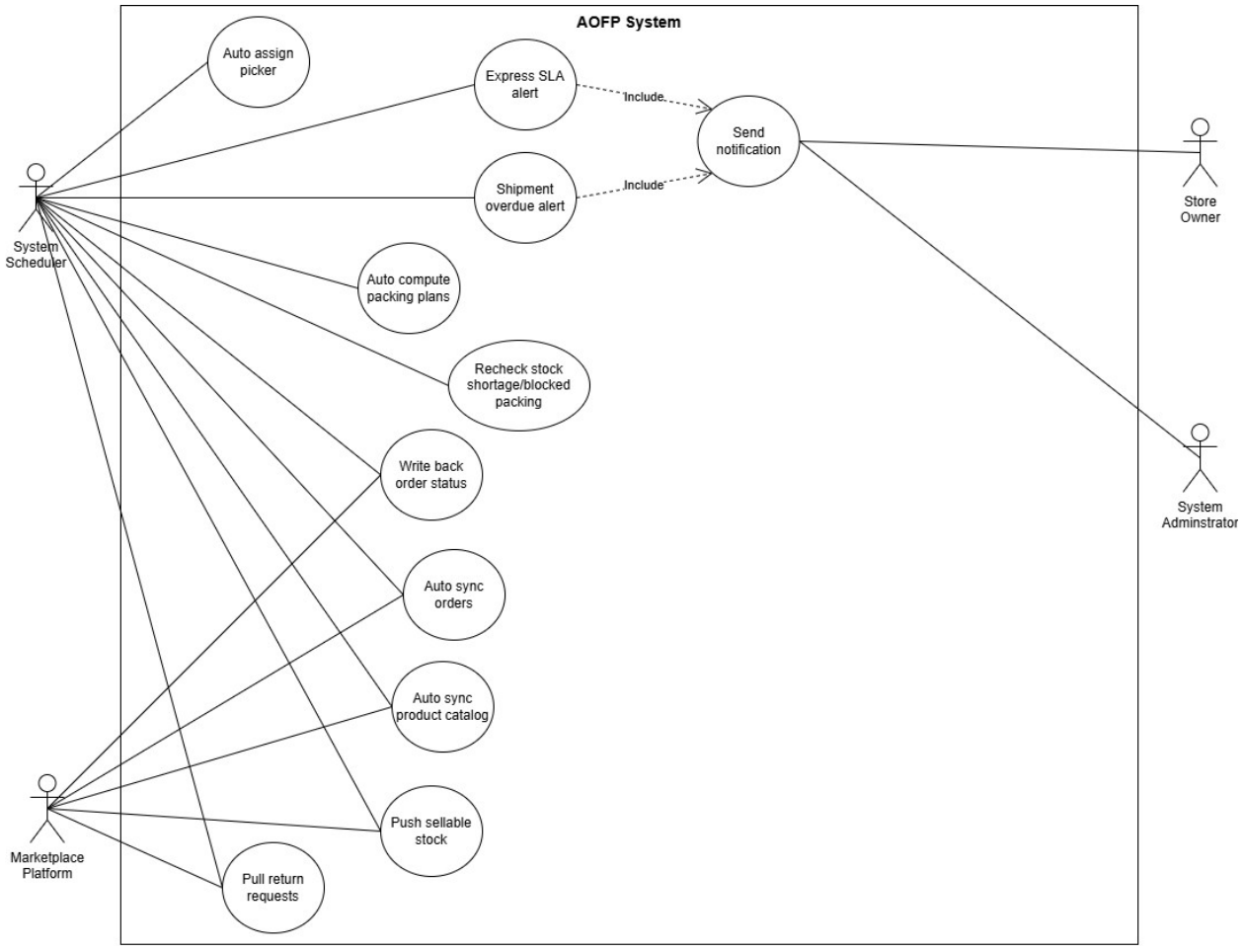


Figure 7. Use case diagram – System Scheduler: background jobs and alerts (UC-52)

2.2.2 Descriptions

ID	Use Case	Actors	Use Case Description
UC-01	Log in	All staff roles	Log in with e-mail and password or with Google (e-mail must already exist). Failed attempts are counted; a new account must change its temporary password before using the portal.
UC-02	Verify two-factor authentication	All staff roles (opt-in)	For an account that has enabled two-factor authentication: enter a TOTP code or a one-time backup code; a device can be trusted for 30 days.
UC-03	Reset forgotten password	All staff roles	Request a reset link by e-mail and set a new password before the link expires.
UC-04	Change password	All staff roles	Change password (forced on first login); all sessions of the account are signed out.
UC-05	Manage user accounts	Admin (Store Owner: view)	Create staff accounts with a role, edit, deactivate / reactivate, reset password, disable two-factor authentication, delete.
UC-06	Set staff leave	Admin, Store Owner	Mark a staff member on leave (until a date); staff on leave are not auto-assigned work.
UC-07	View notifications	All staff roles	View, read and mark all notifications as read (missing items, shortages, plans to approve, failed delivery…).
UC-08	View action history	All staff roles	View history filtered by the categories allowed for the role, view the timeline of one order group; Admin and Store Owner export CSV.
UC-09	Connect marketplace shop	Admin, Lazada / AURELLE	Authorize a shop through OAuth; tokens are stored encrypted.
UC-10	Disconnect marketplace shop	Admin	Deactivate a connected shop; it is no longer synchronized.
UC-11	Configure order sync and consolidation	Admin (Store Owner: view)	Choose order statuses to import, consolidation key fields, time window and statuses of groups that still accept late orders; every change is versioned.
UC-12	Synchronize orders	Admin, System Scheduler	Import new and changed orders on demand or automatically.
UC-13	View omnichannel orders	Admin, Store Owner	List and view orders of all connected channels with items and recipient.
UC-14	View and search order groups	Admin, Store Owner, operational staff	Filter groups by status, priority, platform; search by group id or marketplace order number.
UC-15	Set express priority	Store Owner, Admin	Mark a group as express with a packaging deadline; the system warns before the deadline.
UC-16	Link order groups	Shipping Coordinator, Admin	Link groups of the same recipient (e.g. masked Lazada data) so they are picked and shipped together.
UC-17	Split or merge order groups	Store Owner, Shipping Coordinator, Admin	Split an order out of a consolidated group or merge two groups of the same customer, before any item is picked.
UC-18	Cancel AURELLE order as seller	Store Owner, Admin	Cancel an AURELLE order that cannot be fulfilled; the customer is informed with the reason.
UC-19	Assign warehouse staff	Admin, System Scheduler	Assign a picker to a group manually or automatically (least workload, not on leave).
UC-20	Manage warehouse structure	Admin	Create and edit warehouses, zones, racks and bins with capacity; deactivate or delete when empty.
UC-21	Manage internal SKUs and mappings	Admin	Manage categories, colours and internal SKUs (code rule CATEGORY-MODEL-COLOUR-SIZE), map

ID	Use Case	Actors	Use Case Description
			marketplace seller SKUs to internal SKUs so their stock is pooled.
UC-22	Confirm packaging profile	Warehouse Staff, Admin	Sync the marketplace catalog and confirm package dimensions, weight, fragility, item category and folding of each SKU.
UC-23	Manage bin stock	Warehouse Staff, Admin	Assign a SKU to a bin, add stock, adjust with reason, transfer between bins, stocktake, view movements and lots.
UC-24	Receive stock by lot	Warehouse Staff, Admin	Create a stock receipt (lot code, received date, supplier, lines); stock is consumed oldest lot first.
UC-25	Print labels and documents	Warehouse Staff, Packaging Staff, Shipping Coordinator, Store Owner, Admin	Print bin labels, SKU labels, picking list, stock receipt, packing slip, shipping label and trip manifest (PDF).
UC-26	Pick items	Warehouse Staff	Scan items of an assigned group bin by bin and confirm picking completed.
UC-27	Report missing items	Warehouse Staff	Report a SKU that cannot be found; the group waits for a decision.
UC-28	Decide partial shipment	Packaging Staff, Admin	Ship the available items (customers are informed of short-shipped items) or restart picking.
UC-29	Review and approve packing plan	Packaging Staff, Admin	View the 3D plan per order and approve it to start packing.
UC-30	Reject packing plan	Packaging Staff, Admin	Reject with a reason code, then recompute, pack manually or send the group back to picking.
UC-31	Pack parcels	Packaging Staff, Warehouse Staff	Scan items into each parcel, change box if needed, seal and weigh; view the packing guide.
UC-32	Review held parcel	Packaging Staff, Admin	Accept, reweigh or reopen a parcel whose weight is outside tolerance (different person from the sealer).
UC-33	Handle packing issue	Packaging Staff, Warehouse Staff, Admin	Report damaged / missing / wrong items, replace from shelf, unseal, unpack parcels of cancelled orders.
UC-34	Configure packing and view reports	Store Owner, Admin	Set weight tolerance, fragile cushion, preferred plan, parcel limit, scan requirement; view packing reports.
UC-35	Manage packaging catalog	Admin	Manage boxes, bags and filling materials, stock-in, material rules.
UC-36	Record packaging material movements	Admin, Warehouse Staff (Store Owner: savings)	Record purchases and internal use, view movement history and savings from reused boxes.
UC-37	Create shipment	Shipping Coordinator, Admin	Compare quotes, choose carrier and service and create a shipment (or a same-recipient trip).
UC-38	Fix carrier booking	Shipping Coordinator, Admin	Correct delivery address, re-book parcels without tracking code, cancel a carrier order before pickup.
UC-39	Track delivery	Shipping Coordinator, GHN	Receive carrier status (webhook / sync), record delivered, failed delivery and redelivery.
UC-40	Receive returned parcel	Warehouse Staff	Scan a parcel returned by the carrier; a failed-delivery return ticket is created.
UC-41	Manage carriers and shipping settings	Admin, Store Owner	Manage carriers / services and sender information printed on labels.
UC-42	Review return request	Store Owner, Admin	Approve, reject (reason required) or cancel a return ticket that never arrived.
UC-43	Inspect returned items	Warehouse Staff, Admin	Receive returned goods and decide restock / quarantine / discard per SKU; resolve quarantine later.

ID	Use Case	Actors	Use Case Description
UC-44	Create replacement order	System, Store Owner, Admin	Create the EXC replacement order of an exchange after inspection (retry if it failed).
UC-45	Register and log in (customer)	Customer	Create an AURELLE account and log in.
UC-46	Browse products and manage cart	Customer	Browse categories and products, add variants to cart, update quantities.
UC-47	Place order	Customer	Check out with a shipping address (province + ward) and payment method (COD or bank transfer).
UC-48	Track and cancel order	Customer	Follow the 7-step order progress and delivery events; cancel while the order is not yet packed.
UC-49	Request return or exchange	Customer	Request return & refund, refund only or exchange within 15 days of delivery; cancel a pending request.
UC-50	Manage storefront catalog	AURELLE Web Admin	Manage categories, products, variants (SKU by OptiPack rule) and images.
UC-51	Manage developer apps	AURELLE Web Admin	Create Open Platform apps, rotate secret, revoke / reactivate keys.
UC-52	Run scheduled jobs	System Scheduler	Order sync, packing-plan computation, status write-back, stock push, return sync, SLA and backfill checks.

3. Functional Requirements

3.1 System Functional Overview

OptiPack AI has two web front ends served by one back end: the OptiPack staff portal (React) and the AURELLE website with its customer, web admin and developer pages (Next.js). The AURELLE Open API used as the second marketplace runs as a separate simulator process. Staff roles see only the screens allowed for them (section 3.1.3). Main business workflows WF-01 to WF-06 are described with activity diagrams in Report 4 (SDD), section "Main Business Workflows".

3.1.1 Screens Flow

The screens of the OptiPack staff portal and the AURELLE website, with the roles that use them, are listed in section 3.1.2 Screen Descriptions and section 3.1.3 Screen Authorization.

3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	Common	Login	E-mail / password and Google login, two-factor step when enabled for the account, link to forgot password.
2	Common	Forgot / Reset Password	Request a reset link and set a new password from the e-mailed link.
3	Common	Change Password	Forced change of the temporary password on first login.
4	Common	Profile & Settings	Personal information, password change, optional two-factor setup.
5	Common	Notifications (top bar)	Unread counter and list; clicking a notification opens the related screen.
6	Dashboard	Dashboard Overview	Home screen of Store Owner / Admin: packing report figures and packaging-material savings come from the back end; revenue-by-channel, finance and staff widgets still show sample data (no back-end aggregate API in this release).
7	Orders	Omnichannel Orders / Order Detail	Orders of all channels with filters; detail shows items, recipient and status history.
8	Orders	Groups & Express	Order groups with status, priority, packer, shortage and blocked flags; detail with orders, link, priority.
9	Orders	Order Sync Settings	Consolidation key fields, imported statuses, time window,

#	Feature	Screen	Description
			change history.
10	Warehouse	Warehouse Picking	Assigned groups, picking list per SKU (bin, oldest lot, other bins), scan, report missing, confirm picked.
11	Warehouse	Warehouse Inventory	Bin stock, assign SKU to bin, add / adjust / transfer stock, stocktake, stock receipts and lots.
12	Warehouse	Admin Warehouse	Warehouses, zones, racks, bins, internal SKUs, categories, colours, SKU mappings, product master.
13	Packing	Packing Station	Plans waiting for approval, 3D viewer, approve / reject, packing session (scan, seal, weigh), held parcels, issues.
14	Packing	Admin Packing Plans	All packing plans with status, proof label and recompute.
15	Packing	Packaging Rules / Packaging Materials	Boxes, bags, filling materials, rules, purchases, internal use and savings.
16	Shipping	Shipping & Fulfillment	Packed groups, quotes, create shipment / trip, carrier parcels and tracking, failed delivery, labels.
17	Returns	Returns	Return tickets (customer requests and failed deliveries): approve / reject, receive, inspect, quarantine, replacement.
18	People	Staff Directory	Staff list, roles, leave status.
19	Admin	Admin Home / Roles	User accounts, roles, two-factor reset, reactivation.
20	Admin	Marketplace Connections	Connect / disconnect Lazada and AURELLE shops, shop status.
21	Admin	AI & Packing Settings / Analytics	Packing settings, packing reports and feedback.
22	AURELLE	Home / Shop / Product Detail	Customer catalog, product with variants and sellable stock.
23	AURELLE	Cart / Checkout	Cart and checkout with new two-level address (province, ward) and payment method.
24	AURELLE	My Orders / Order Detail	Order list and detail with 7-step progress bar, events, cancel and return actions.
25	AURELLE	My Returns / Return Detail	Return and exchange requests with status and exchange shipment progress.
26	AURELLE	Customer Login / Register	Customer account access; Admin web accounts are redirected to the admin page.
27	AURELLE	Web Admin (Products, Categories)	Storefront catalog management with SKU built from OptiPack categories, colours and sizes.
28	AURELLE	Developer Portal	Open Platform apps: app key / secret, rotate, revoke.

3.1.3 Screen Authorization

X = screen available. Back-end endpoints enforce the same roles (role-based guard on every request).

Screen	Admin	Store Owner	Warehouse Staff	Packaging Staff	Shipping Coord.
Dashboard Overview	X	X			
Omnichannel Orders / Order Detail	X	X			
Groups & Express	X	X			
Order Sync Settings (edit)	X	view			
Returns	X	X			
Warehouse Picking	X		X		
Warehouse Inventory	X		X		
Admin Warehouse (structure, SKUs)	X				
Packing Station	X			X	
Packaging Materials	X	X	X	X	

Screen	Admin	Store Owner	Warehouse Staff	Packaging Staff	Shipping Coord.
Packaging Rules	X	X			
Shipping & Fulfillment	X				X
Staff Directory	X	X			
Analytics	X	X			
Admin Home / Roles / Marketplace / Packing Plans / AI	X				
Profile & Settings, Notifications	X	X	X	X	X
Action History (own categories)	X	X	X	X	X
Export history CSV	X	X			

3.1.4 Non-Screen Functions

#	Feature	System Function	Description
1	Orders	Marketplace order auto-sync	Every minute: for each connected Lazada / AURELLE shop whose sync interval has passed (default 5 minutes, configurable 5–1,440), import new / changed orders page by page, consolidate and create groups.
2	Orders	Marketplace webhooks	Receive AURELLE webhooks (order changed, product updated, return requested / cancelled), verify HMAC signature, ignore duplicates.
3	Orders	Order-group backfill	Every 5 minutes: create missing groups and resume groups stuck in awaiting_packaging.
4	Orders	Express SLA check	Every 10 minutes: warn when an express group is less than 1 hour from its deadline and flag overdue groups.
5	Catalog	Product master sync	Hourly catalog sync and daily full sync at 03:00 (Asia/Ho_Chi_Minh); auto-map and auto-create internal SKUs that follow the code rule.
6	Packing	Packing-plan auto compute	Every 10 seconds: compute plans for picked groups (BRKGA solver, optional CP-SAT proof service).
7	Packing	Packing-blocked recheck	Every minute: retry groups blocked by unmeasured SKUs after the SKU is confirmed.
8	Packing	Rejected plan reminder	Every 15 minutes: remind once when a rejected plan passes its 2-working-hour deadline.
9	Shipping	Carrier webhook	Receive GHN status callbacks, map to shipment status, ignore duplicates.
10	Shipping	Shipment SLA scan	Every 15 minutes: mark overdue deliveries.
11	Write-back	Order / return status write-back	Every 15 seconds: send queued status changes to AURELLE, retry with back-off (up to 8 attempts).
12	Write-back	Stock push	Every minute: push available stock of mapped SKUs to AURELLE.
13	Returns	Marketplace returns sync	Every 2 minutes: pull AURELLE return requests and cancellations not received by webhook.
14	Common	Notification and e-mail	Create in-app notifications by role / user; send account e-mails through SMTP.

3.1.5 Entity Relationship Diagram

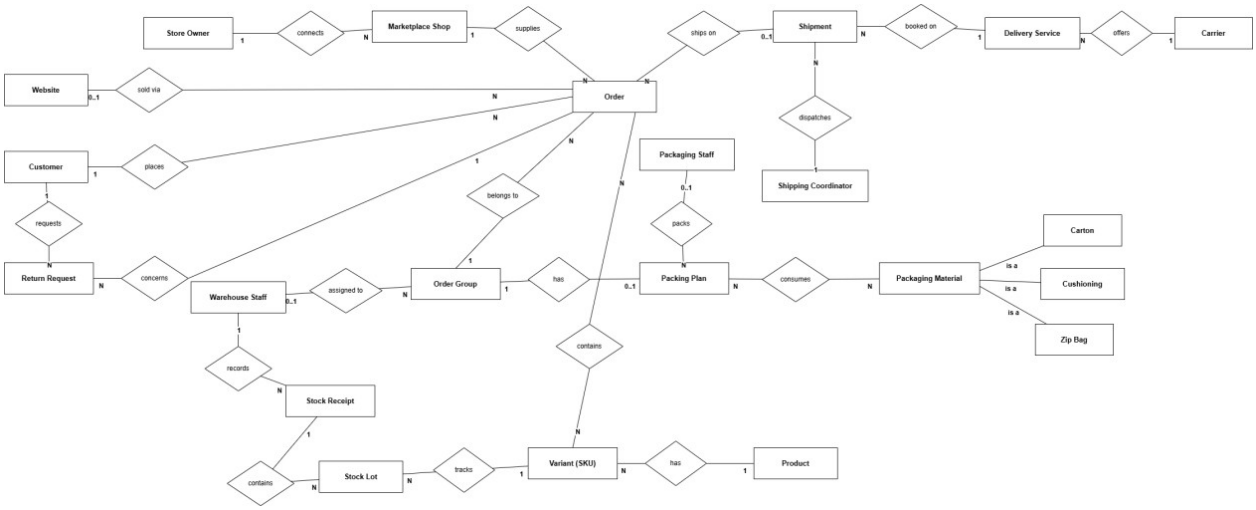


Figure 8. Logical entity relationship diagram

Entities Description

#	Entity	Description
1	User	Staff account with one role (Admin, Store Owner, Warehouse Staff, Packaging Staff, Shipping Coordinator), lockout, two-factor and leave data.
2	Marketplace Shop	A connected shop of Lazada / AURELLE (encrypted tokens, connection status).
3	Order	Marketplace order with recipient, items, statuses and consolidation key.
4	Order Group	Unit of fulfilment: one or more orders of the same recipient that are picked, packed and shipped together.
5	Stock Reservation	Quantity of a SKU reserved for an order group in a warehouse.
6	Pick Event	One scan of a SKU while picking (quantity, bin, lots, pick round).
7	Warehouse / Zone / Bin Location	Physical structure of the warehouse with bin capacity.
8	SKU Bin Assignment	Stock of one SKU in one bin.
9	Stock Lot / Stock Receipt	Lot received on a date (FIFO) and the receipt that created it.
10	Inventory Movement	Ledger entry of every stock change (pick, adjust, transfer, restock…).
11	Category / Colour / Master SKU	Internal product classification and internal SKU code.
12	Marketplace SKU Mapping	Link from a marketplace seller SKU to an internal SKU (pooled stock).
13	Product Master	Marketplace catalog item with confirmed packaging profile (dimensions, weight, fragility, category).
14	Packing Plan	Active 3D packing plan of a group: orders, parcels, boxes, placements, scans, session.
15	Packaging Material / Movement	Boxes, bags and filling materials with new / reused stock and their movements.
16	Shipment / Shipment Event	Outbound shipment of a group with carrier parcels and its status history.
17	Return Request	Return, refund-only, exchange or failed-delivery ticket with inspection results.
18	Notification / Activity Log	In-app notification and business action log used by history screens.
19	Status Outbox	Queue of status changes to be written back to AURELLE.
20	Storefront Customer / Product / Variant / Order / Return	Entities of the AURELLE website.

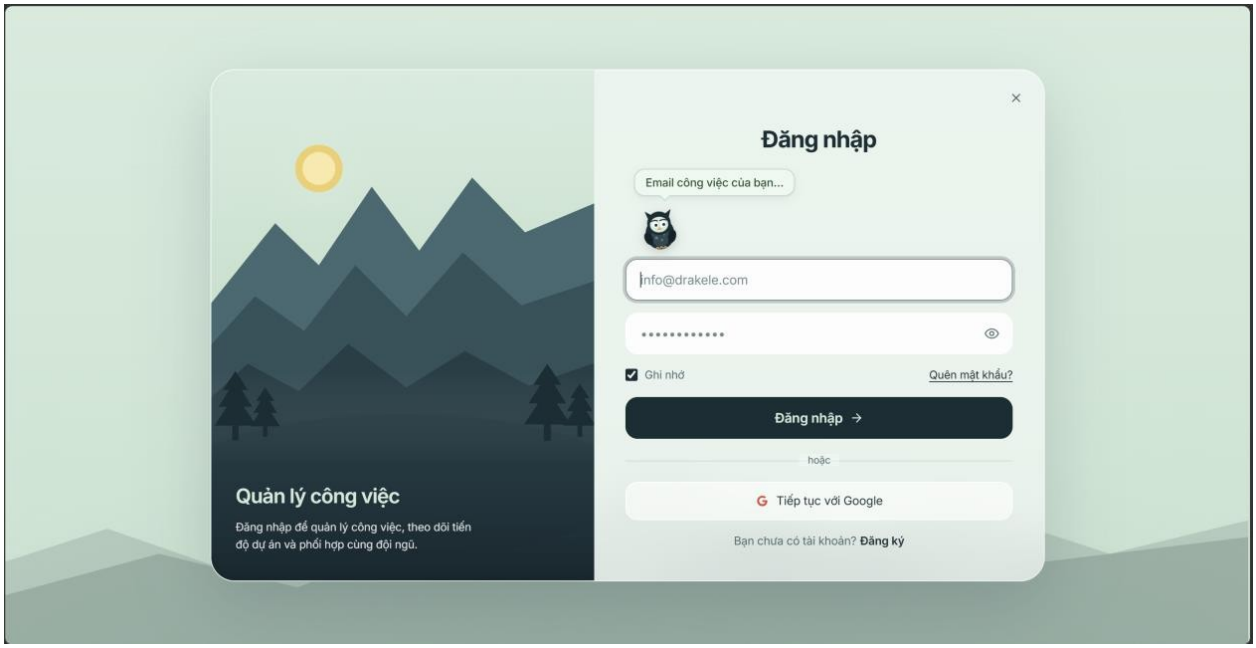
3.2 Authentication and Account Management

3.2.1 Log in

Function trigger: Login screen; Google button; /login.

Function description: All staff roles. Authenticates the user and opens the home screen of the role.

Screen layout:



Function details:

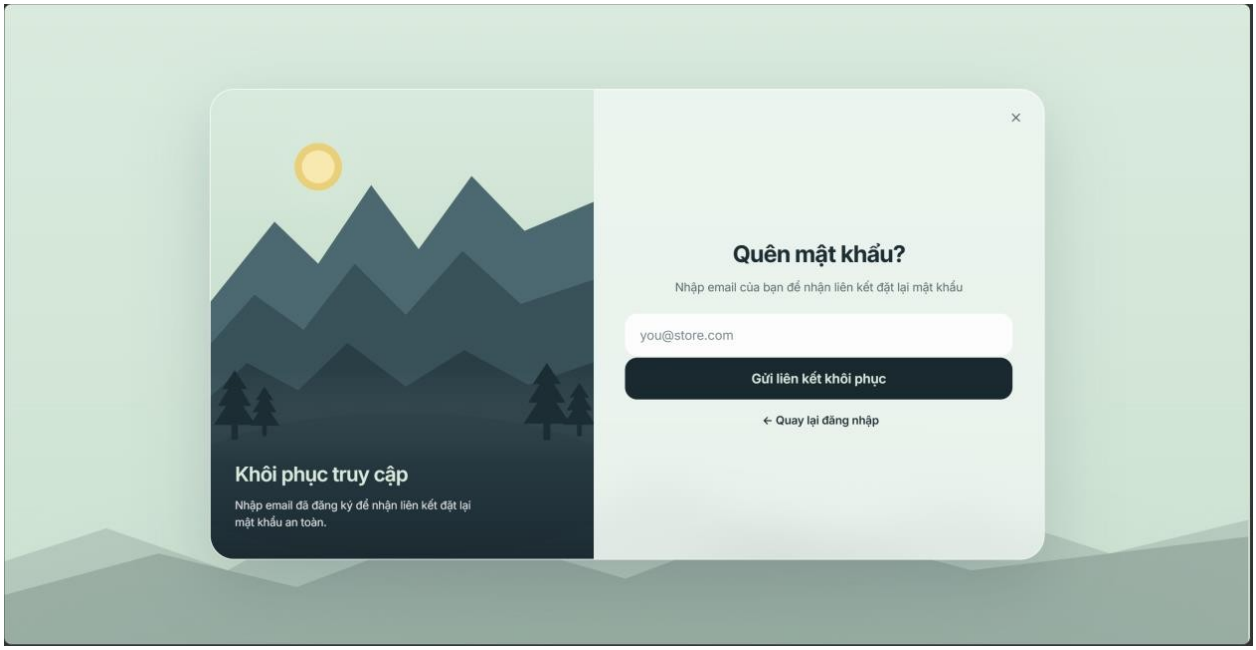
- E-mail and password are required; a wrong password increases the failed-attempt counter. After 5 consecutive failures the account is locked for 15 minutes.
- Google login succeeds only when the verified Google e-mail already belongs to an active account.
- A user who has enabled two-factor authentication must enter a TOTP code or a backup code unless the device is trusted (30 days); two-factor authentication is optional for every role.
- An account created by Admin has a temporary password that must be changed within 72 hours; otherwise the account is locked until Admin reactivates it.
- Access token lifetime 24 hours, refresh token 30 days with rotation; reuse of an old refresh token revokes the session.

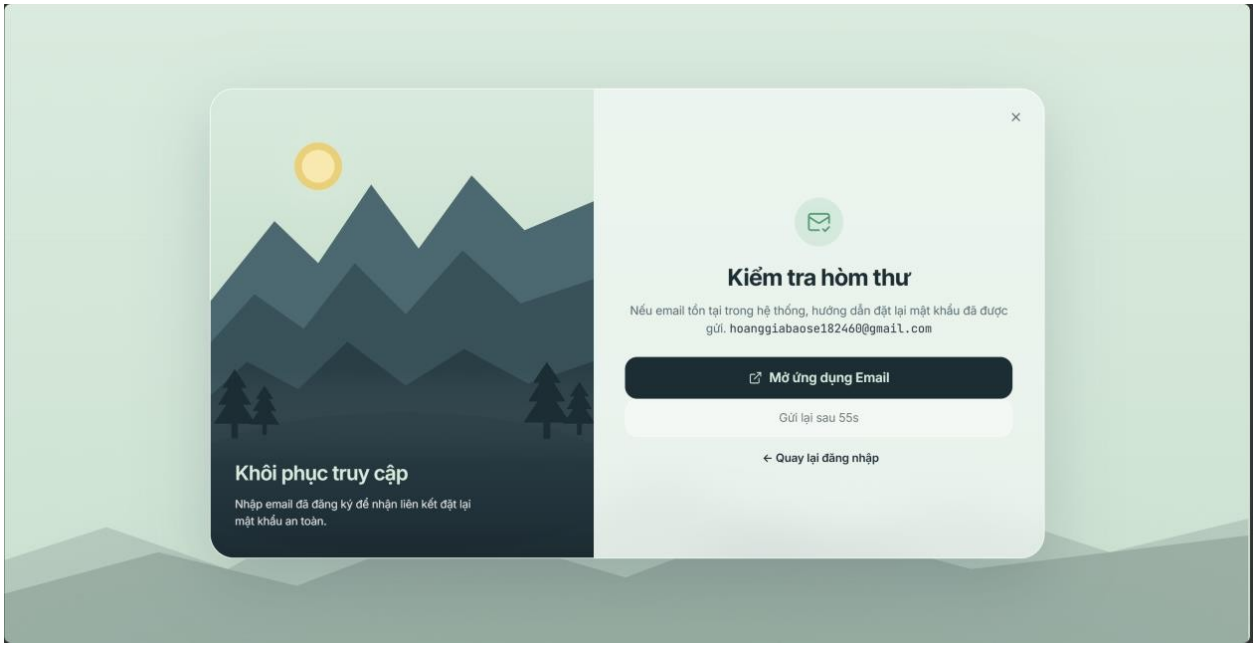
3.2.2 Reset and change password

Function trigger: Forgot Password link; Change Password screen.

Function description: All staff roles. Recovers access or replaces the password.

Screen layout:





Function details:

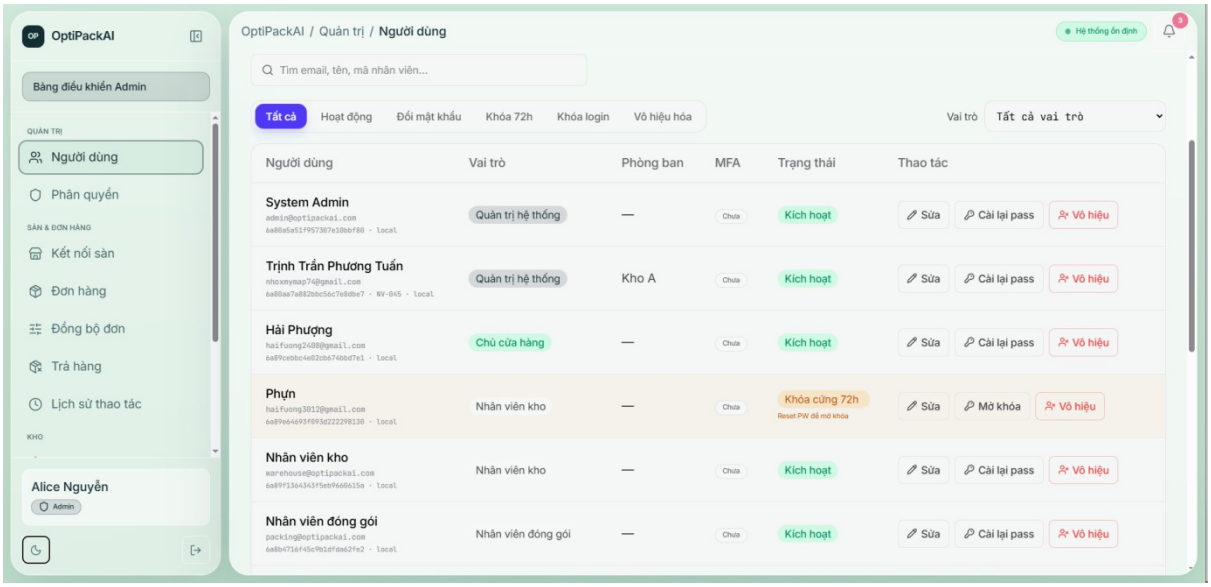
- The reset e-mail always shows the same generic message whether the e-mail exists or not.
- The reset link expires after 30 minutes and can be used once.
- New password: at least 8 characters with upper case, lower case, digit and special character.
- Changing or resetting the password signs out all sessions of the account.

3.2.3 Manage user accounts and leave

Function trigger: Admin Home / Roles; Staff Directory.

Function description: Admin manages accounts; Store Owner views staff and sets leave.

Screen layout:



Function details:

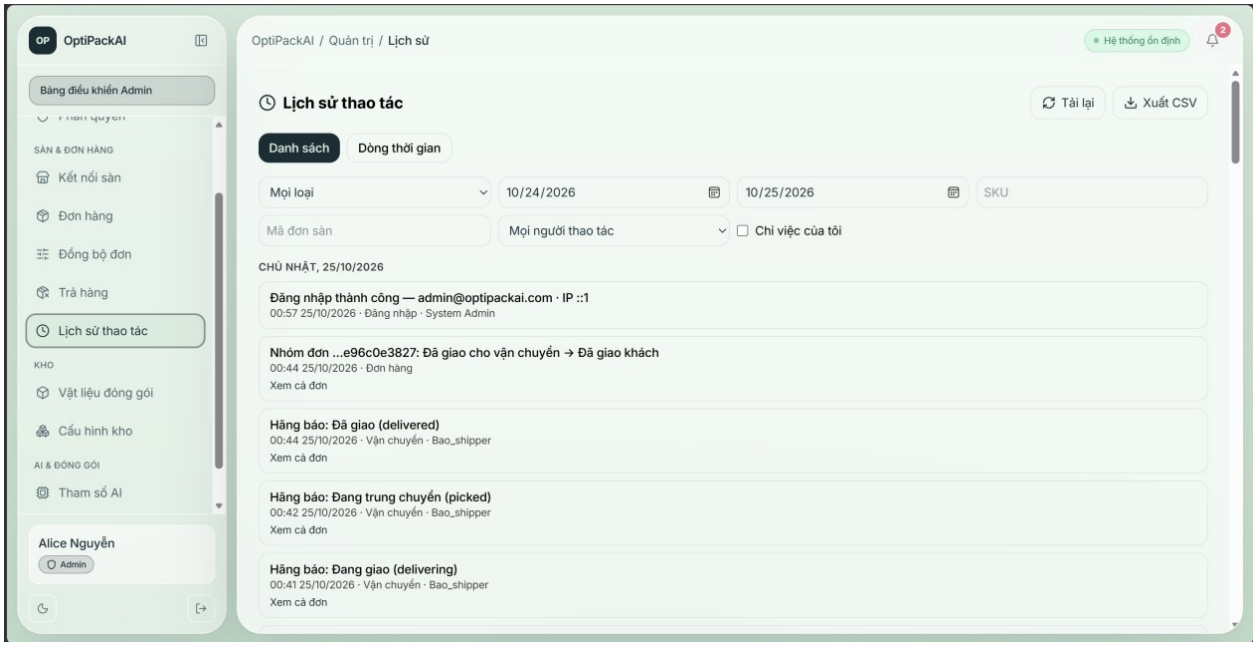
- Admin creates the account and the system e-mails a temporary password.
- Deactivating an account revokes its sessions; reactivation clears lock state.
- Staff on leave (until a date) are skipped by auto-assignment and cannot be assigned manually.

3.2.4 Action history

Function trigger: History screen; Timeline tab of an order group.

Function description: All roles; each role sees only its categories. Admin and Store Owner export CSV (max 5,000 rows).

Screen layout:



Function details:

- Categories: orders, picking, packing, shipping, returns, warehouse, packaging materials, login (Admin only).

- Default period 30 days; filter by date, SKU, order code, actor, "my actions".

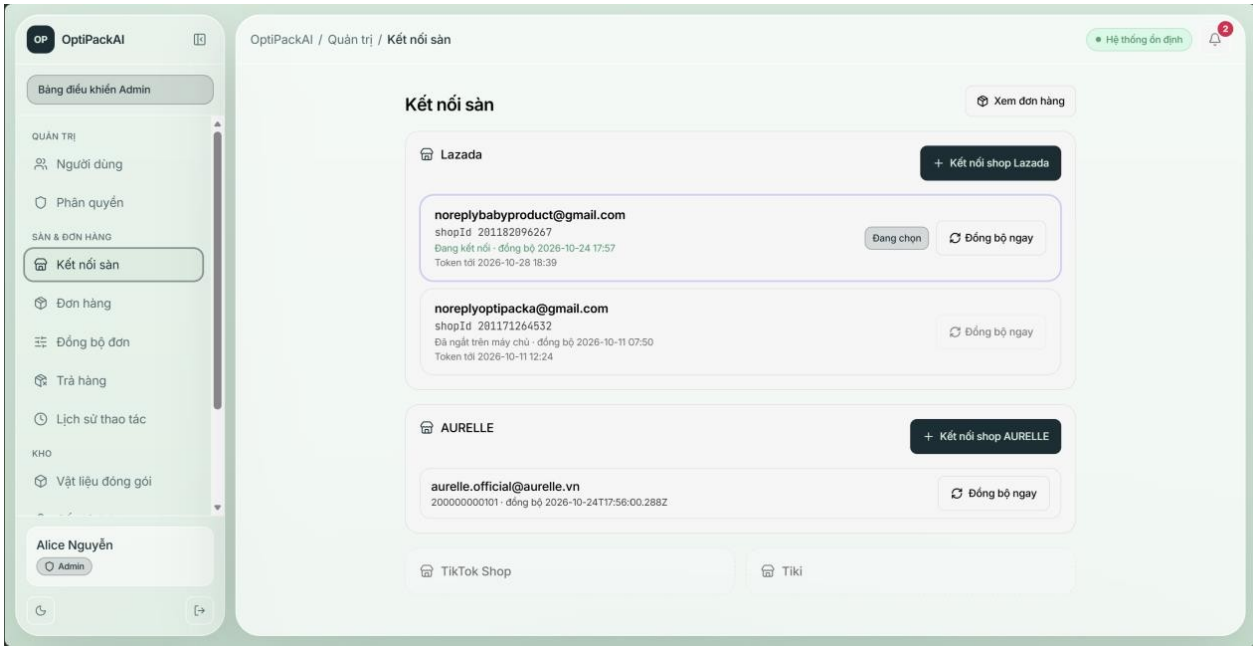
3.3 Marketplace Integration and Order Intake

3.3.1 Connect / disconnect shop

Function trigger: Admin → Marketplace Connections.

Function description: Admin authorizes a Lazada or AURELLE shop through OAuth.

Screen layout:



Function details:

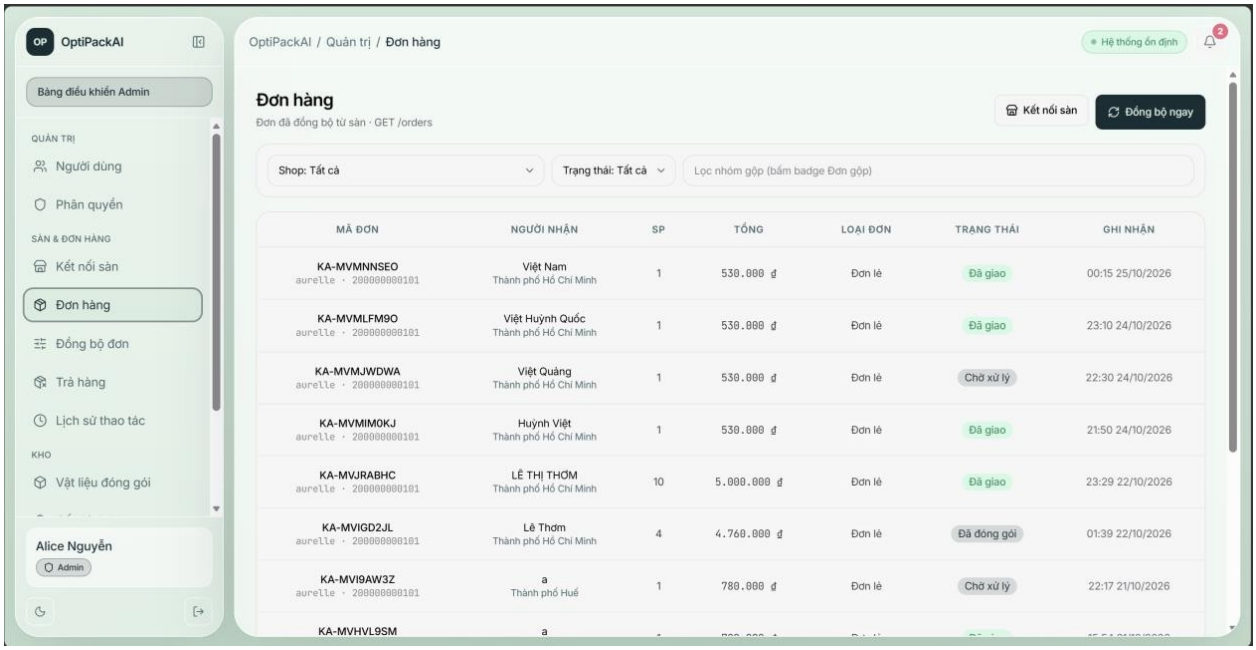
- Tokens are encrypted (AES-256-GCM) before storage and refreshed automatically.
- A disconnected shop is no longer synchronized; reconnecting reactivates it.

3.3.2 Synchronize and consolidate orders

Function trigger: Automatic (scheduler every minute, each shop at its interval, default 5 minutes), webhook (AURELLE), or "Sync" button (Admin).

Function description: System imports orders and builds order groups.

Screen layout:



Function details:

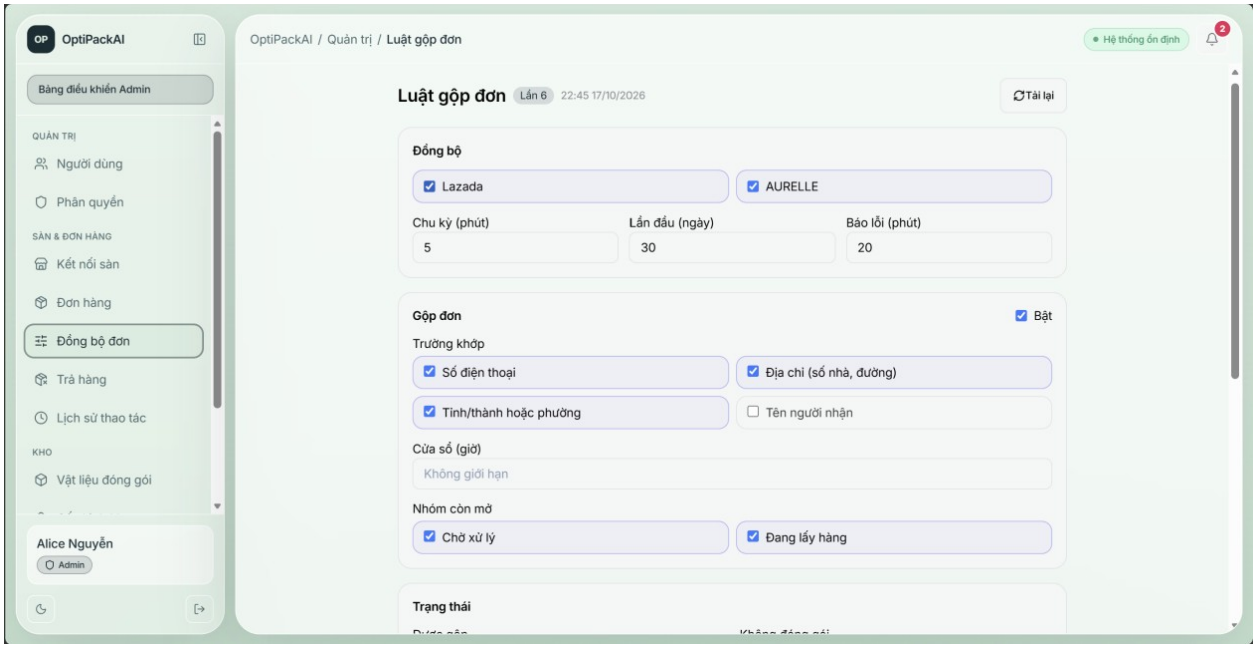
- Orders with the same consolidation key (default: platform + phone + address + city), same shop and an open group join that group; otherwise a new group is created.
- A new group is reserved stock, assigned the least-busy Warehouse Staff and moved to picking.
- If available stock does not cover the group, it is flagged stock shortage and the Store Owner is notified.
- If recipient names in a group differ, Store Owner, Coordinator and the picker are notified.
- A cancelled order re-checks its group; a group without fulfillable orders is cancelled automatically.

3.3.3 Order sync settings

Function trigger: Order Sync Settings screen.

Function description: Admin edits, Store Owner views.

Screen layout:



Function details:

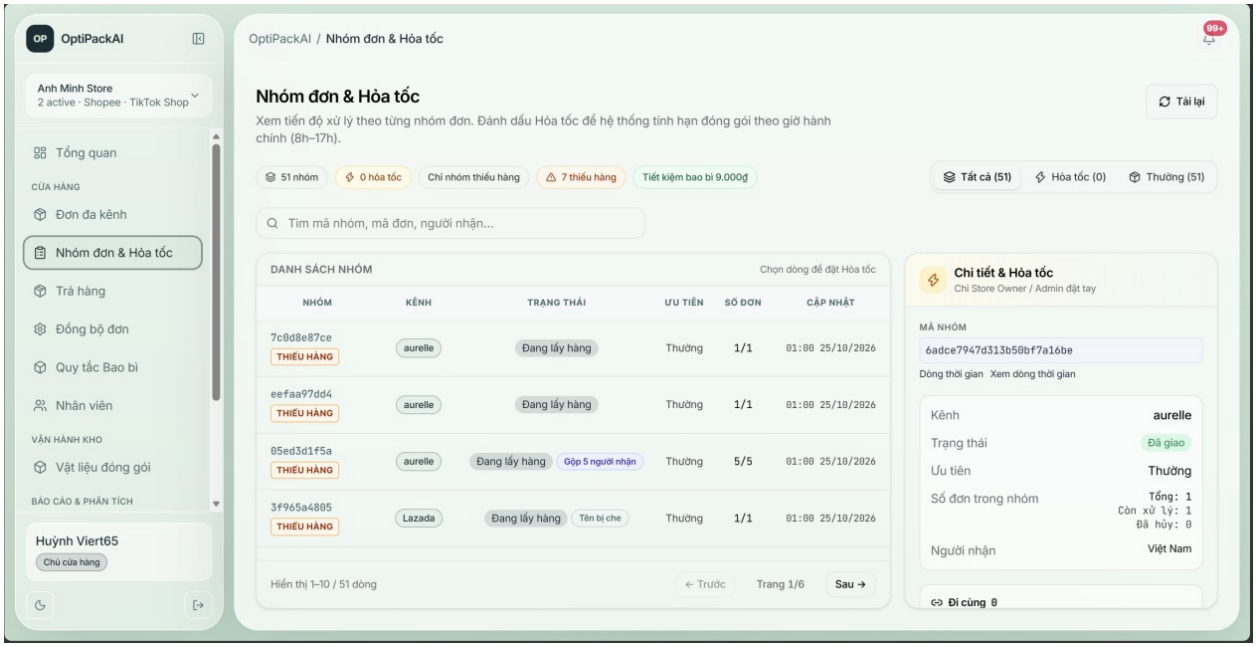
- Every save creates a new version with author and time; history can be viewed.

3.3.4 Manage order groups

Function trigger: Groups & Express screen.

Function description: Store Owner, Shipping Coordinator, Admin.

Screen layout:



Function details:

- Express priority sets a packaging deadline in business hours; warning when less than 1 hour remains.
- Split / merge only while the group is awaiting_packaging or picking with no item scanned; merge requires same shop and same real phone number; a reason is mandatory.

- Seller cancel is allowed only for AURELLE orders not yet handed to the carrier; the customer is refunded if paid.

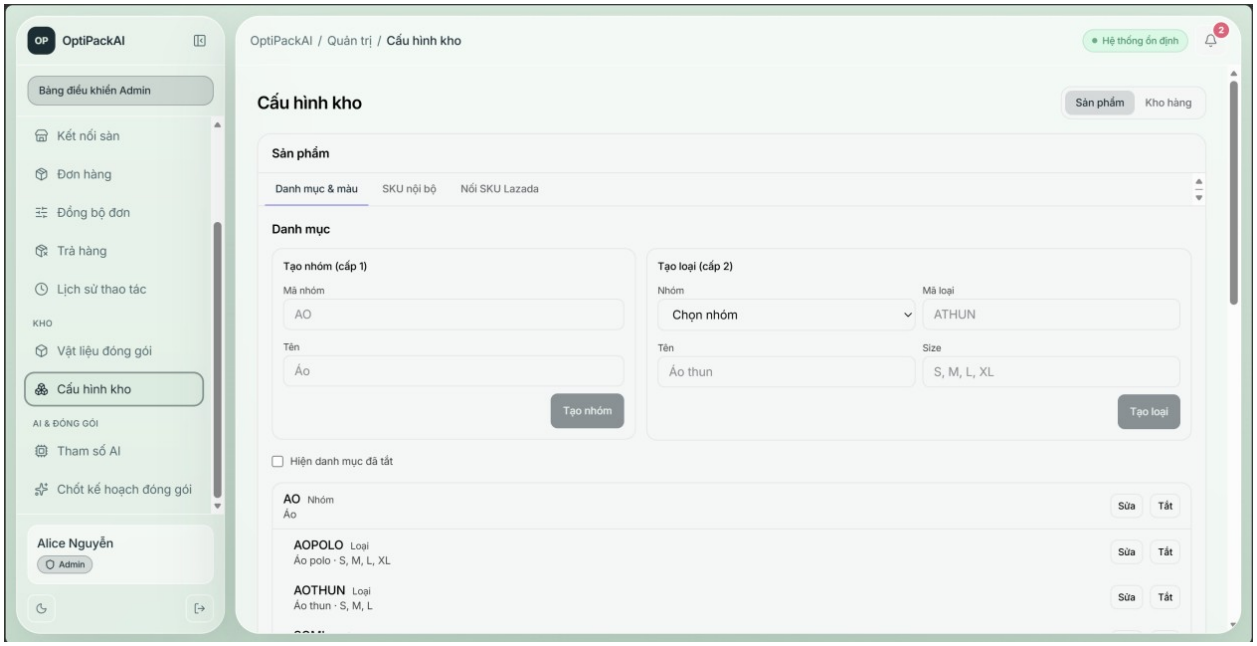
3.4 Warehouse and Inventory

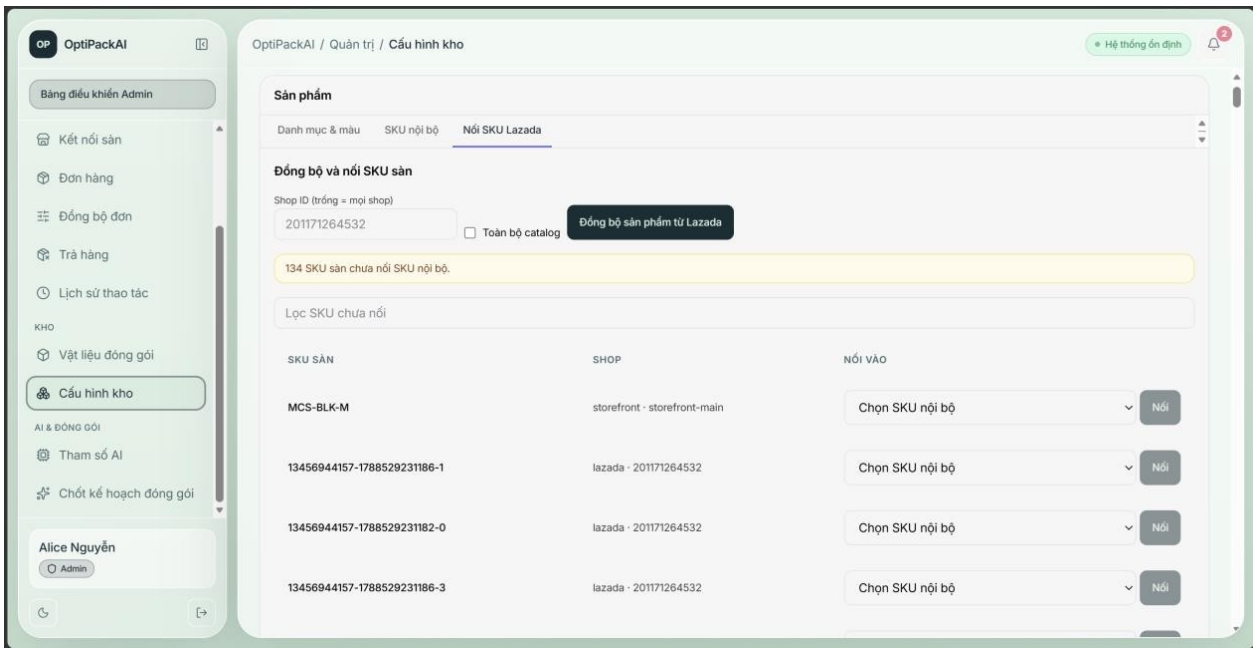
3.4.1 Warehouse structure

Function trigger: Admin Warehouse screen.

Function description: Admin creates warehouses, zones, racks and bins.

Screen layout:





Function details:

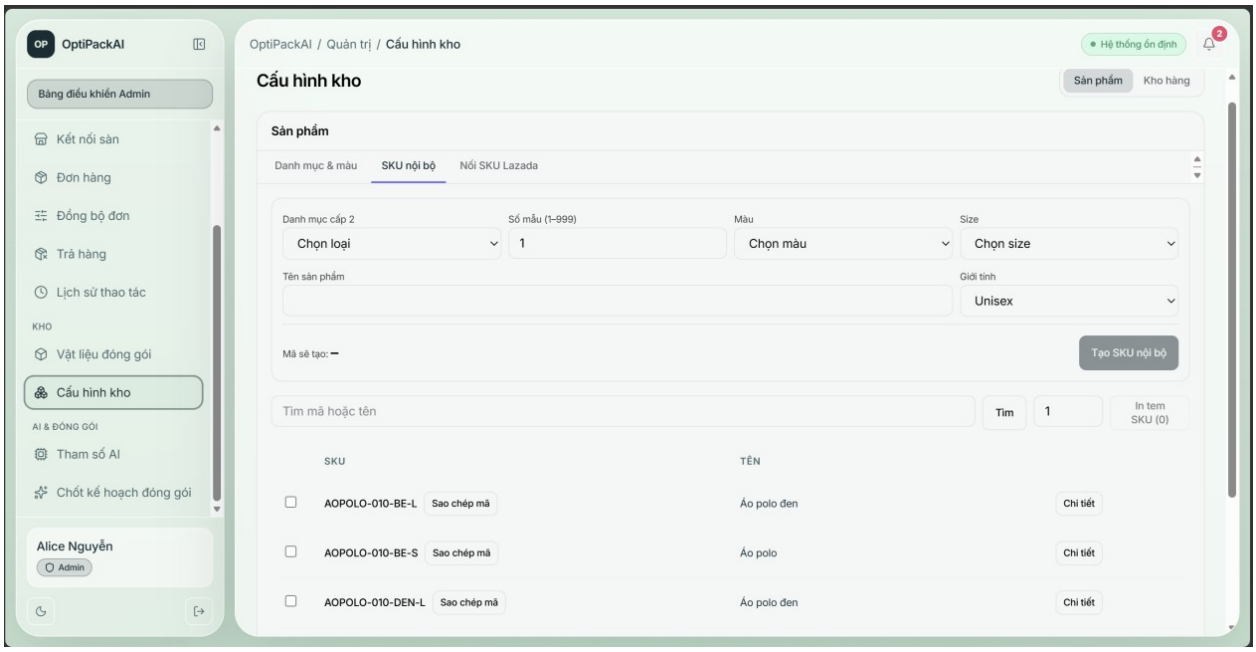
- Bin codes are generated by rack; a bin can have a capacity in units.
- A warehouse, zone or bin can be deleted permanently only when it holds no stock.

3.4.2 Internal SKU and mapping

Function trigger: Admin Warehouse → SKU tab.

Function description: Admin manages internal SKUs and seller-SKU mappings.

Screen layout:



Function details:

- Internal SKU code follows CATEGORY-MODEL-COLOUR-SIZE; category and colour must exist.
- Mapping a seller SKU to an internal SKU pools the stock and moves existing reservations.
- Seller SKUs with the same code as an internal SKU are mapped automatically.

3.4.3 Packaging profile

Function trigger: Product master list → profile form.

Function description: Warehouse Staff confirms dimensions, weight, fragility, item category and folding.

Screen layout:

[Insert screenshot of the "Packaging profile" screen]

Function details:

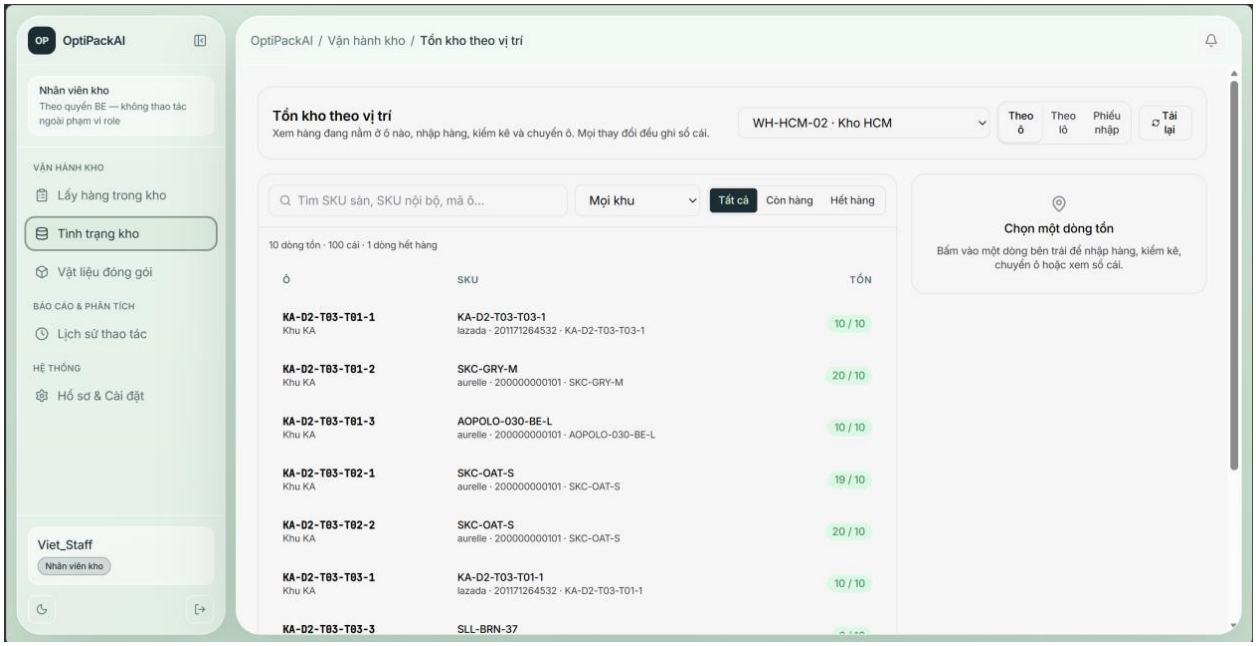
- Only the six clothing categories can be folded in half.
- A group whose SKU has no confirmed profile cannot get a packing plan (packing-blocked).

3.4.4 Bin stock and stock receipt

Function trigger: Warehouse Inventory screen.

Function description: Warehouse Staff assigns SKUs to bins and receives stock by lot.

Screen layout:



Function details:

- Putting a SKU into a second bin of the same warehouse requires explicit confirmation.
- A receipt that exceeds bin capacity is rejected unless forced.
- Each receipt line creates a lot; picking consumes the oldest lot first (FIFO).
- Every change writes an inventory movement; stock-in re-checks groups with shortage.

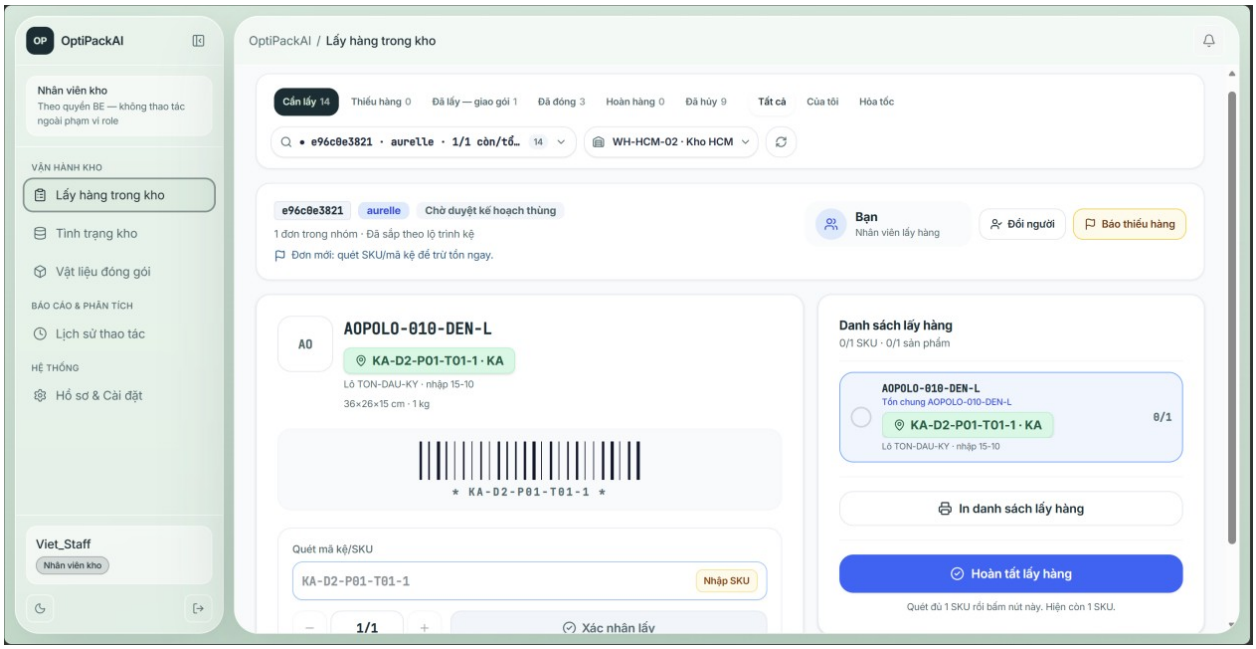
3.5 Picking

3.5.1 Pick items

Function trigger: Warehouse Picking screen → group.

Function description: Warehouse Staff assigned to the group.

Screen layout:



Function details:

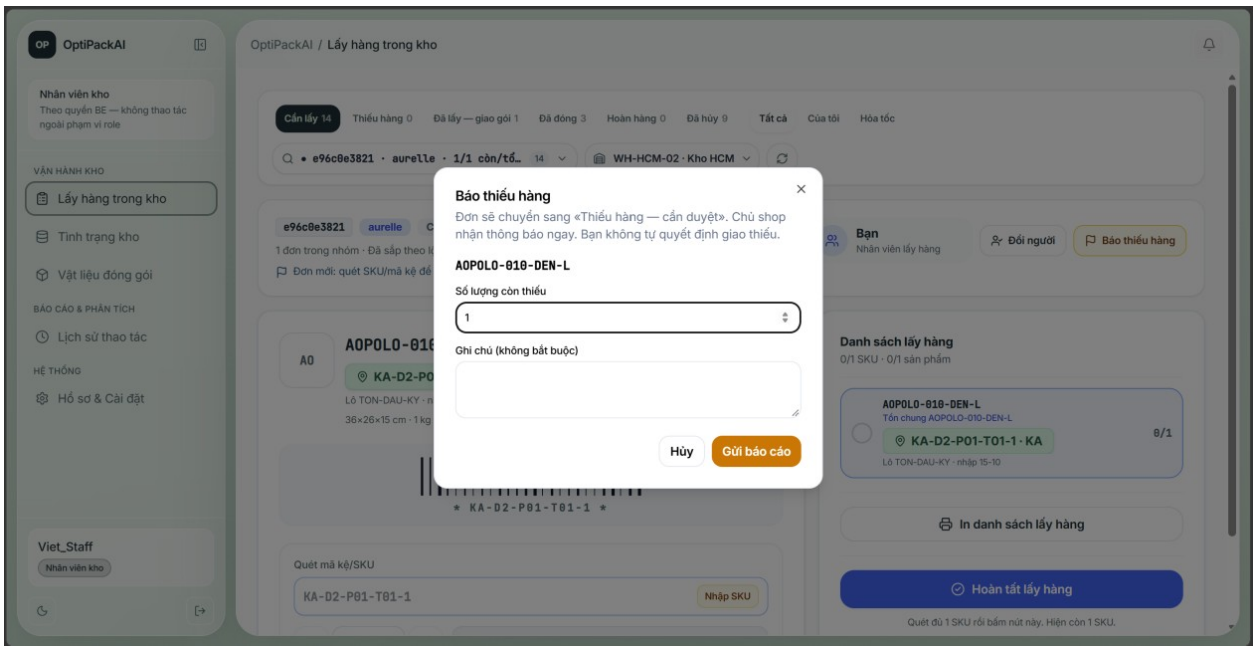
- The picking list shows per SKU: quantity, bin, oldest lot, other bins and split per order for consolidated groups.
- A scan is rejected when the SKU is not in the group, the quantity exceeds the ordered quantity, the warehouse differs or bin stock is insufficient.
- A repeated scan with the same client event id is not counted twice.
- The first scan notifies AURELLE customers "warehouse is preparing your order".
- "Picking completed" is allowed only when every SKU is fully picked.

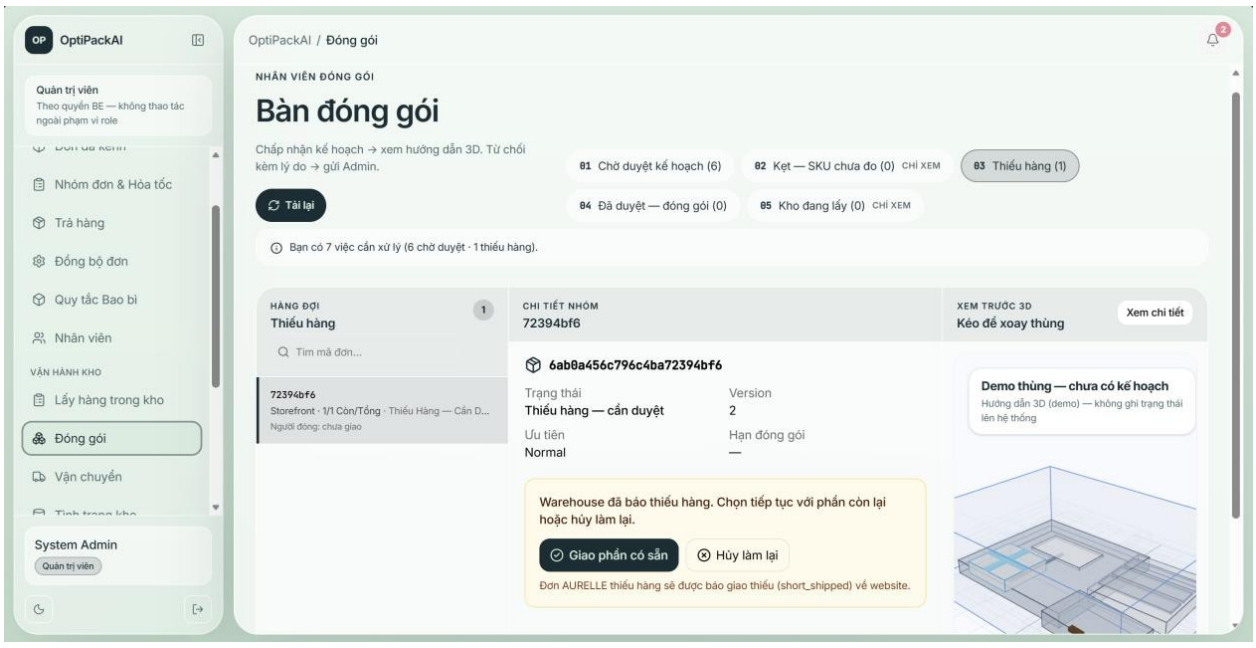
3.5.2 Report missing and decide partial

Function trigger: Picking screen → Report missing; Packing Station → shortage.

Function description: Warehouse Staff reports; Packaging Staff / Admin decides.

Screen layout:





Function details:

- Reporting moves the group to partial_needs_review and notifies Store Owner and Packaging Staff.
- Ship available: group becomes picked, shortage is allocated to oldest orders first and AURELLE customers are informed (not charged).
- Re-pick: a new pick round starts, picked items are returned to their bins and stock is re-reserved.

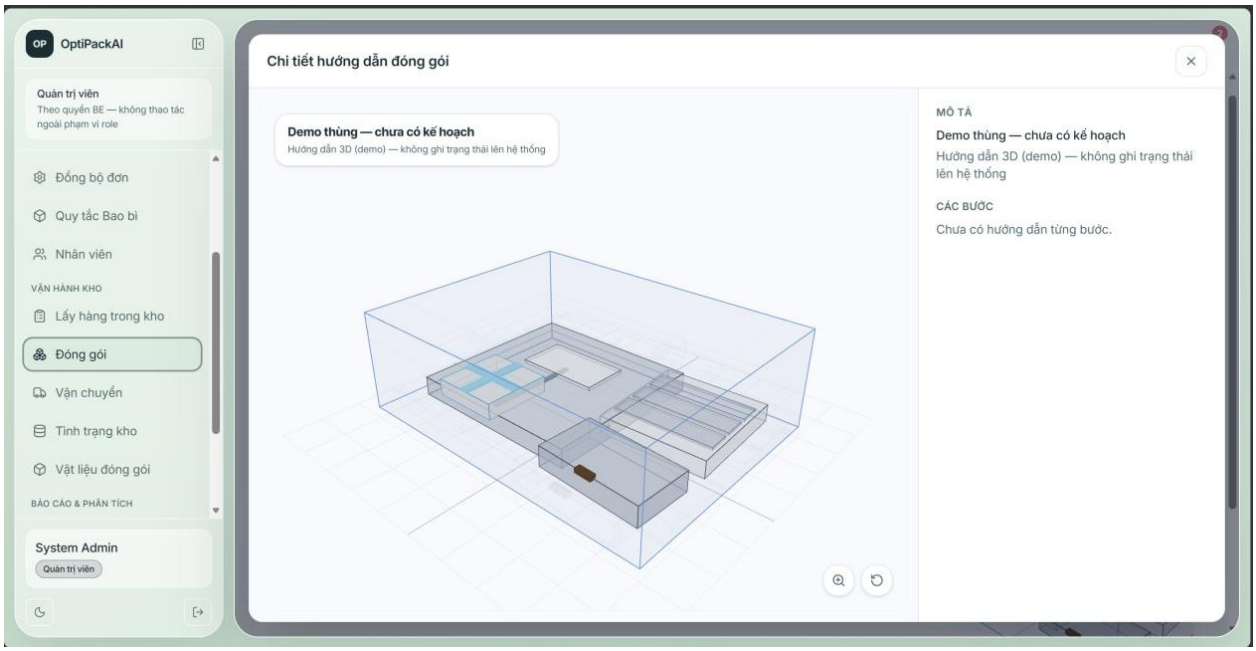
3.6 Packing

3.6.1 Compute and approve packing plan

Function trigger: Automatic; Packing Station.

Function description: System computes; Packaging Staff / Admin approves.

Screen layout:



Function details:

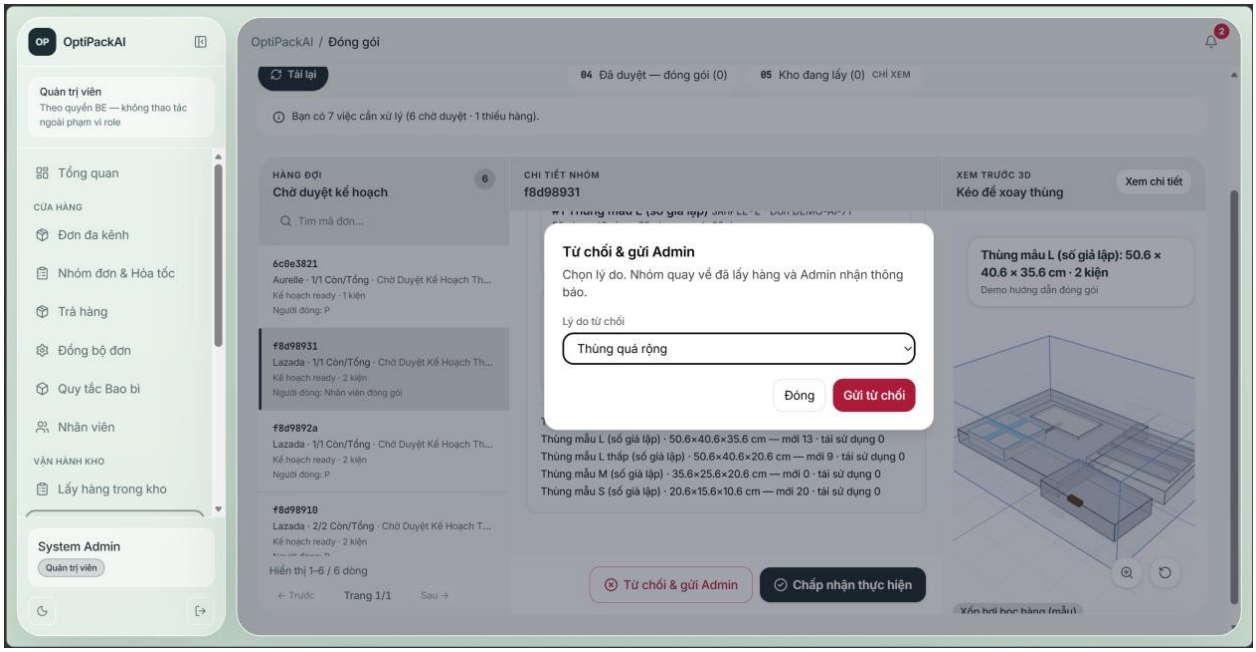
- One plan per order group; each parcel belongs to exactly one order.
- The plan is assigned automatically to the least-busy packer not on leave.
- Approval is possible only when every item of every order is placed.

3.6.2 Reject and recover

Function trigger: Packing Station → Reject.

Function description: Packaging Staff / Admin.

Screen layout:



Function details:

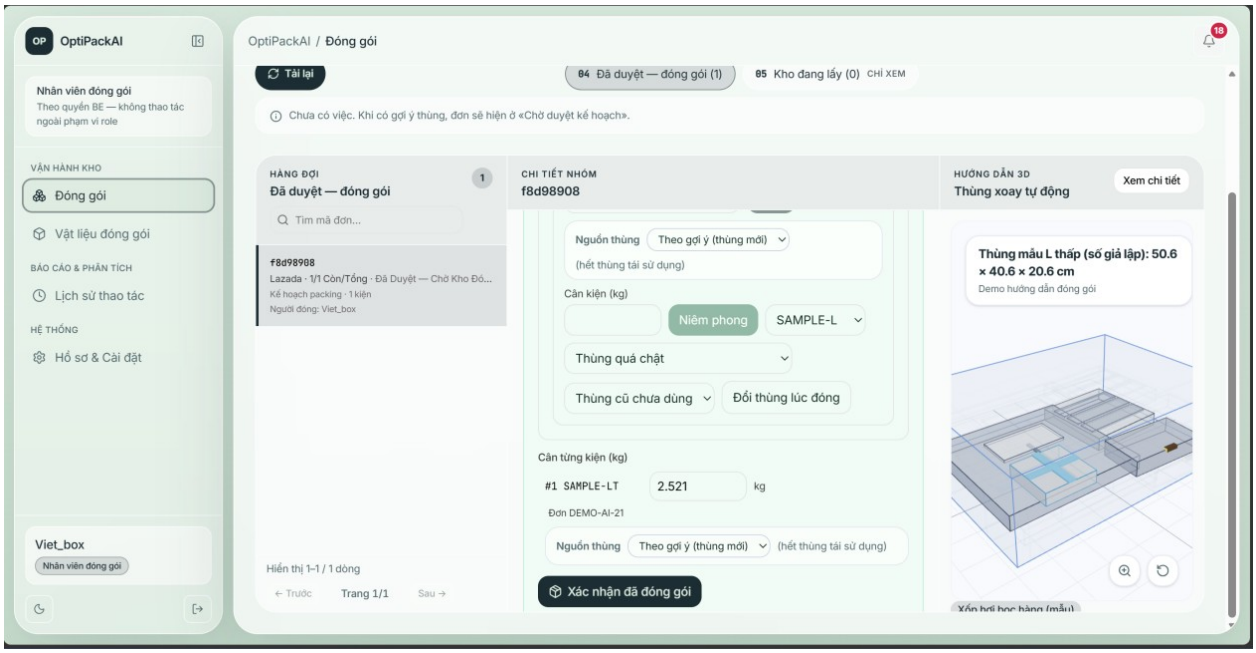
- Reason is chosen from a code list; "Other" needs a note; Admin and Store Owner are notified with a 2-working-hour deadline.
- Recovery: recompute (exclude boxes / prefer cheaper), manual packing approved by another person, or back to picking.

3.6.3 Pack parcels

Function trigger: Packing Station → session.

Function description: Packaging Staff / Warehouse Staff.

Screen layout:



Function details:

- Items must be scanned into the correct parcel; wrong parcel or extra item is blocked.

- Sealing deducts the box (reused shelf first, new box for fragile items) and materials.
- Weight outside tolerance (default 20%) puts the parcel on hold for review by another person.
- "Quick pack" without scanning is blocked when scanning is mandatory or the group has several recipient names.
- The last sealed parcel moves the group to packed and notifies AURELLE customers.

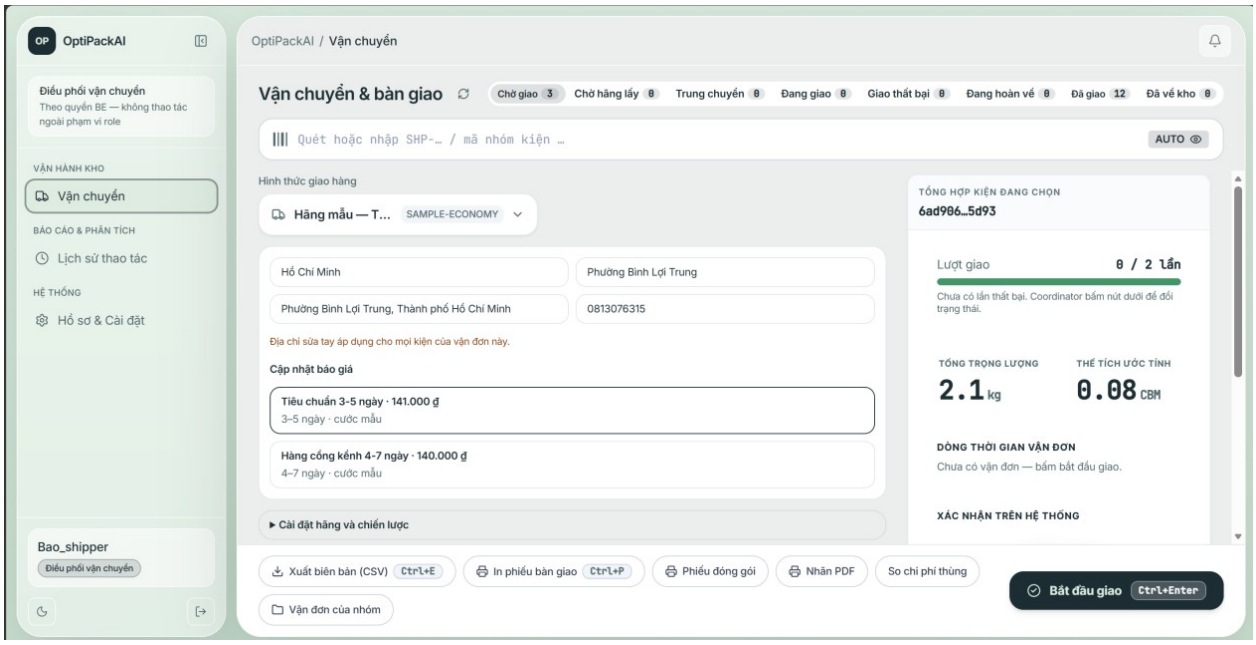
3.7 Shipping

3.7.1 Create shipment

Function trigger: Shipping & Fulfillment screen.

Function description: Shipping Coordinator / Admin.

Screen layout:



Function details:

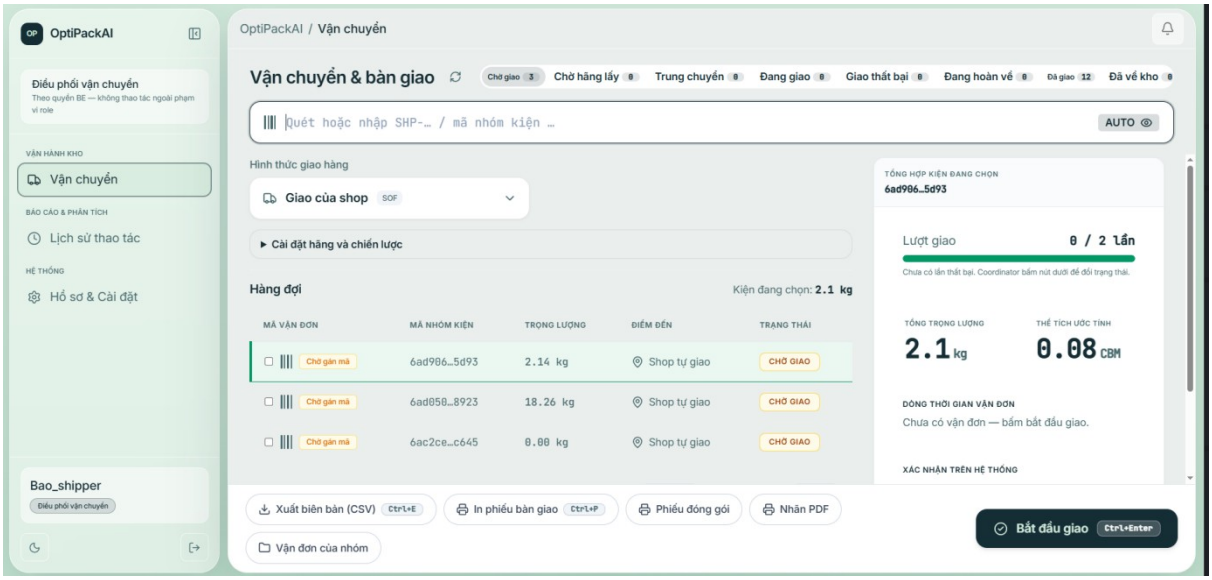
- Only packed groups can be shipped; several groups of the same recipient can be shipped as one trip.
- One GHN order is created per parcel with the recipient of that parcel's order.
- Rejected bookings (e.g. address) are corrected and re-booked without creating duplicates.

3.7.2 Track delivery

Function trigger: Automatic (GHN webhook / sync); Shipping screen.

Function description: GHN, Shipping Coordinator.

Screen layout:



Function details:

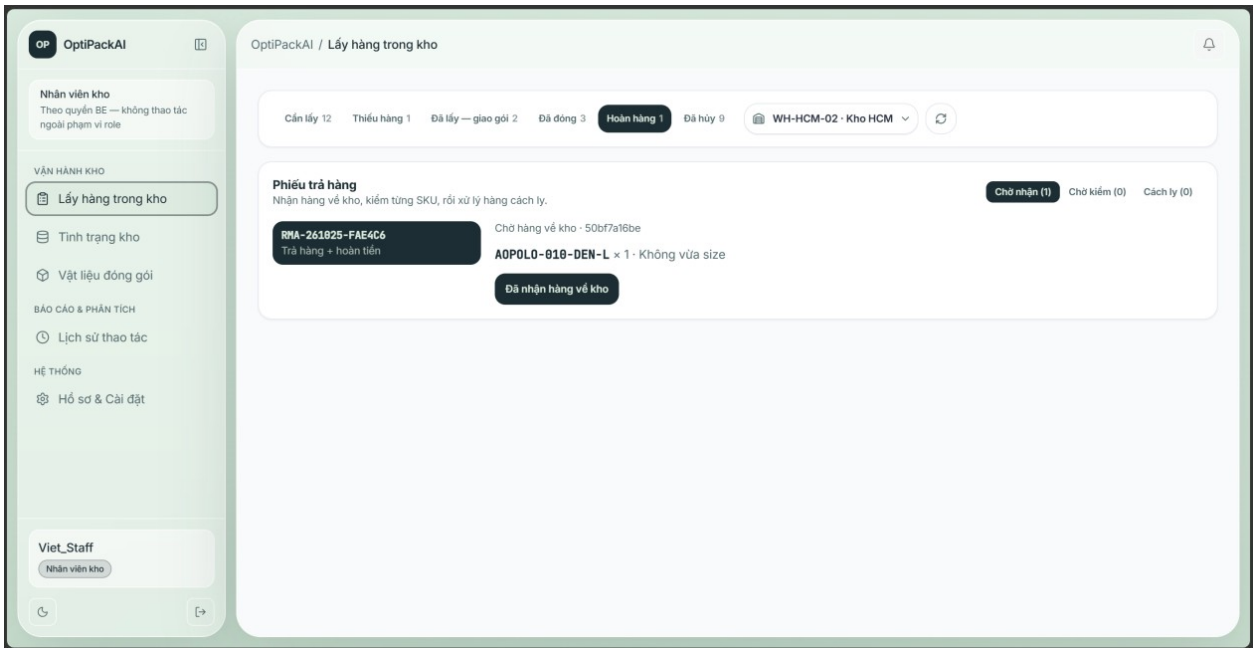
- GHN statuses are mapped to awaiting_pickup, in_transit, out_for_delivery, delivered, delivery_failed, returning_to_warehouse.
- Without carrier API, a second failed attempt or customer refusal automatically returns the shipment.
- Every change is sent to AURELLE so customers see the progress.

3.7.3 Receive returned parcel

Function trigger: Shipping / Warehouse screen → Receive return.

Function description: Warehouse Staff.

Screen layout:



Function details:

- Shipment becomes returned_to_warehouse, group becomes returned and a failed-delivery return ticket is created for inspection.

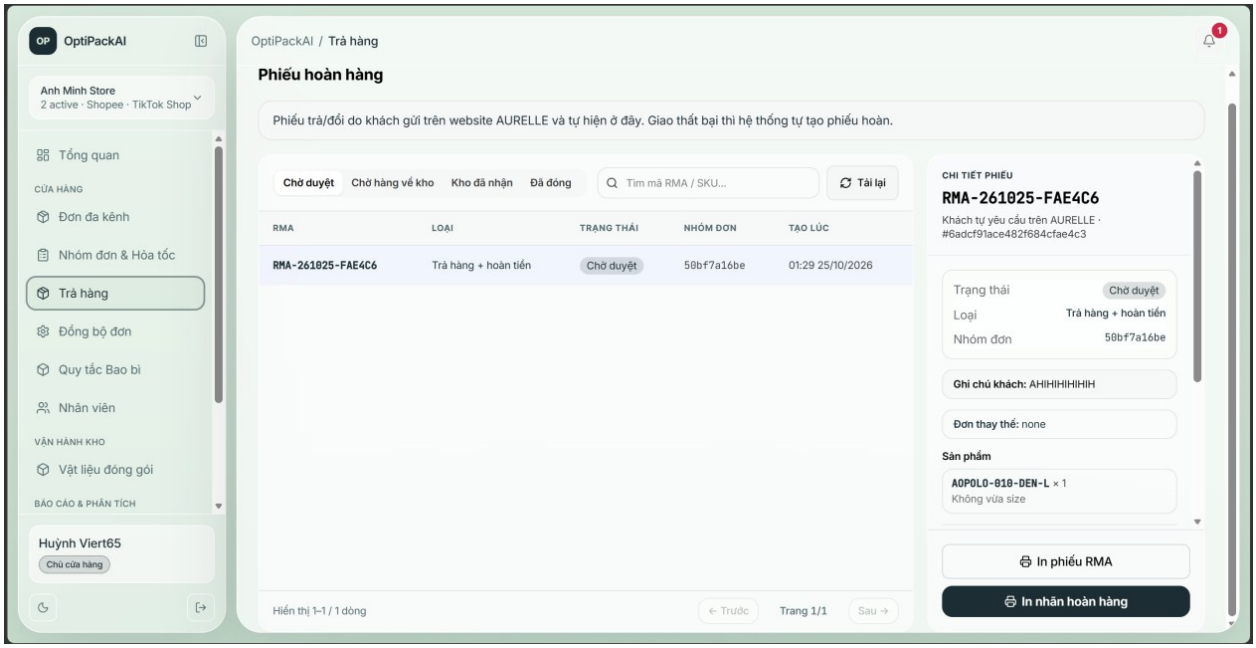
3.8 Returns and Exchanges

3.8.1 Review return ticket

Function trigger: Returns screen.

Function description: Store Owner / Admin.

Screen layout:



Function details:

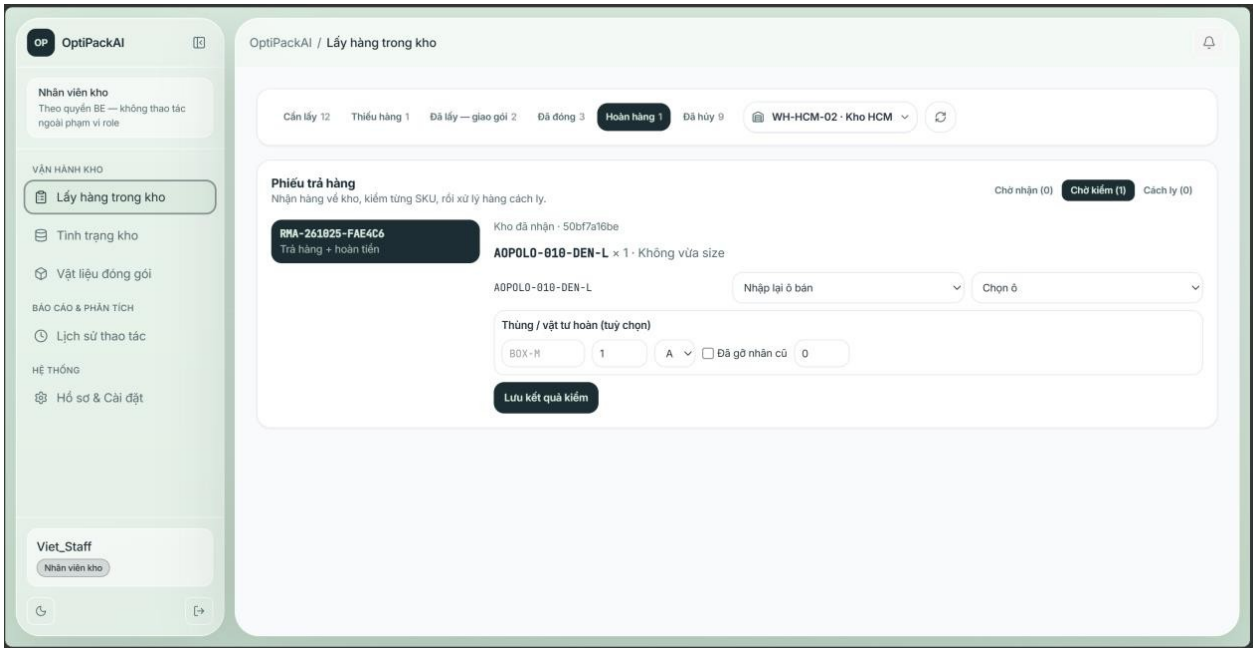
- The creator of a ticket cannot approve it.
- Reject and cancel require a reason.
- Refund-only is closed at approval without goods returned.

3.8.2 Receive and inspect

Function trigger: Returns screen → Receive / Inspect.

Function description: Warehouse Staff / Admin.

Screen layout:



Function details:

- Each SKU is restocked, quarantined or discarded; restocked items return to a bin and write the ledger.
- Exchange tickets create a replacement order EXC-… (price 0) that goes through picking, packing and shipping.

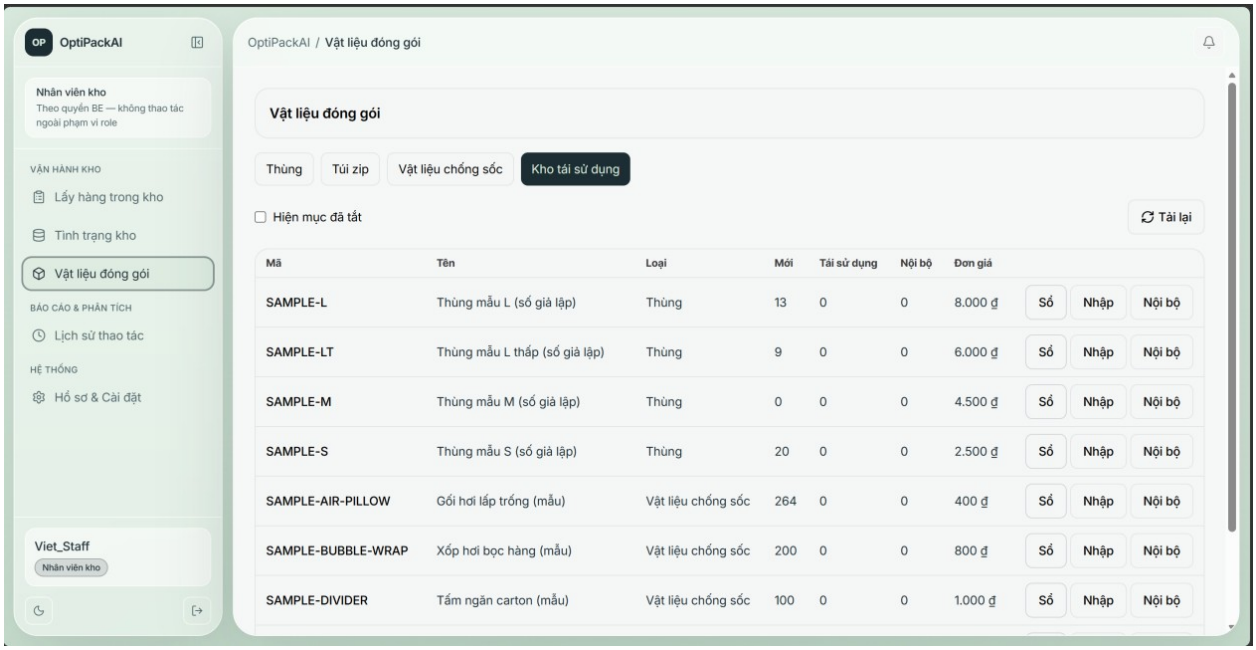
3.9 Packaging Materials

3.9.1 Manage packaging catalog and movements

Function trigger: Packaging Materials / Rules screens.

Function description: Admin manages; Warehouse Staff records stock-in and use; Store Owner views savings.

Screen layout:



Function details:

- Boxes have new and reused stock; fragile items use new boxes unless allowed in settings.
- Every purchase, internal use, consumption and recovery is recorded as a movement.

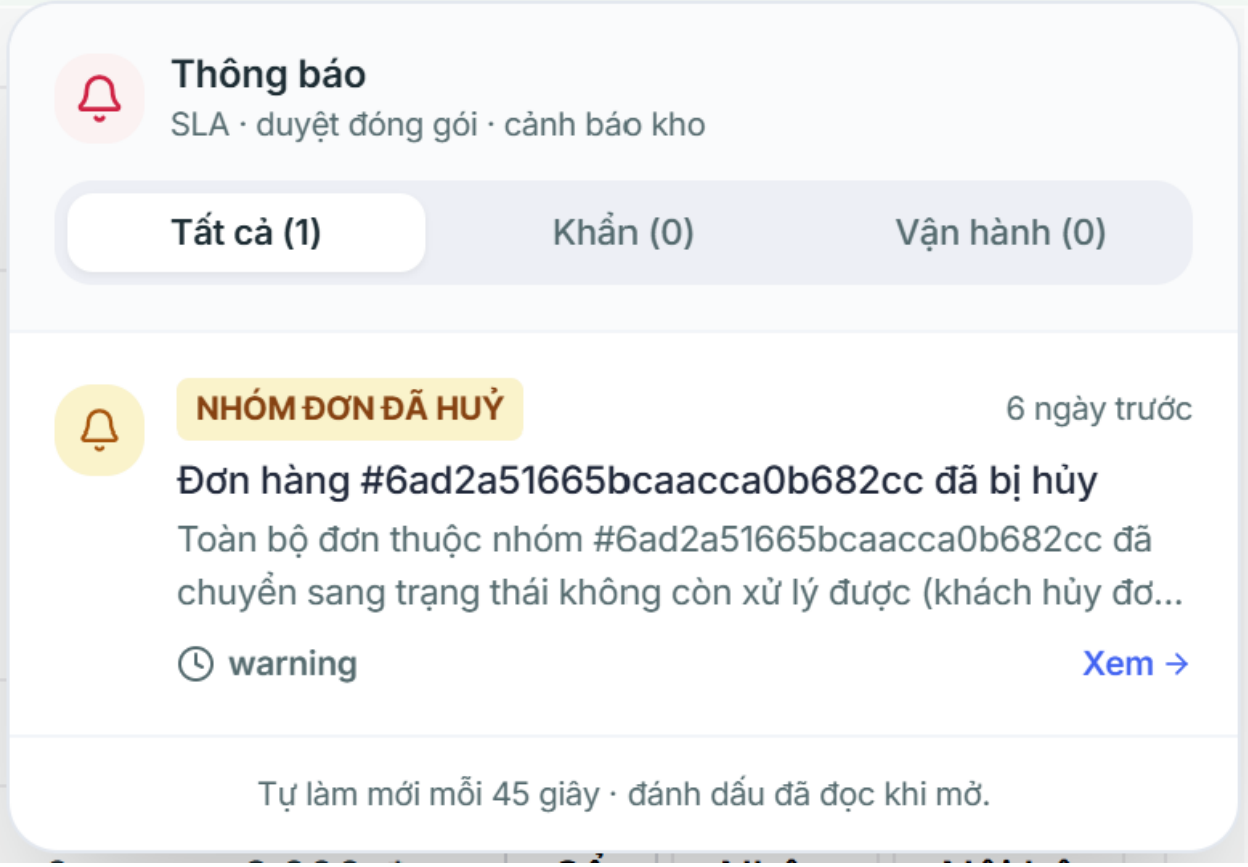
3.10 Notifications

3.10.1 Notifications

Function trigger: Top bar bell.

Function description: All staff roles.

Screen layout:



Function details:

- Notifications are addressed to a role or to a user and link to the related entity.

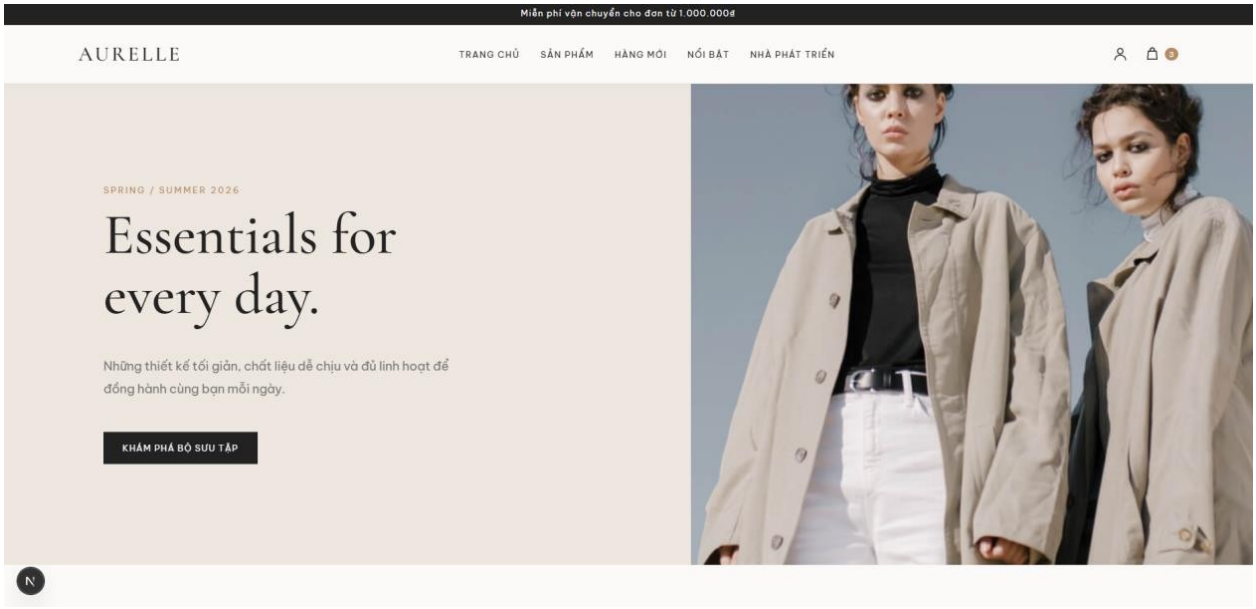
3.11 AURELLE Customer Website

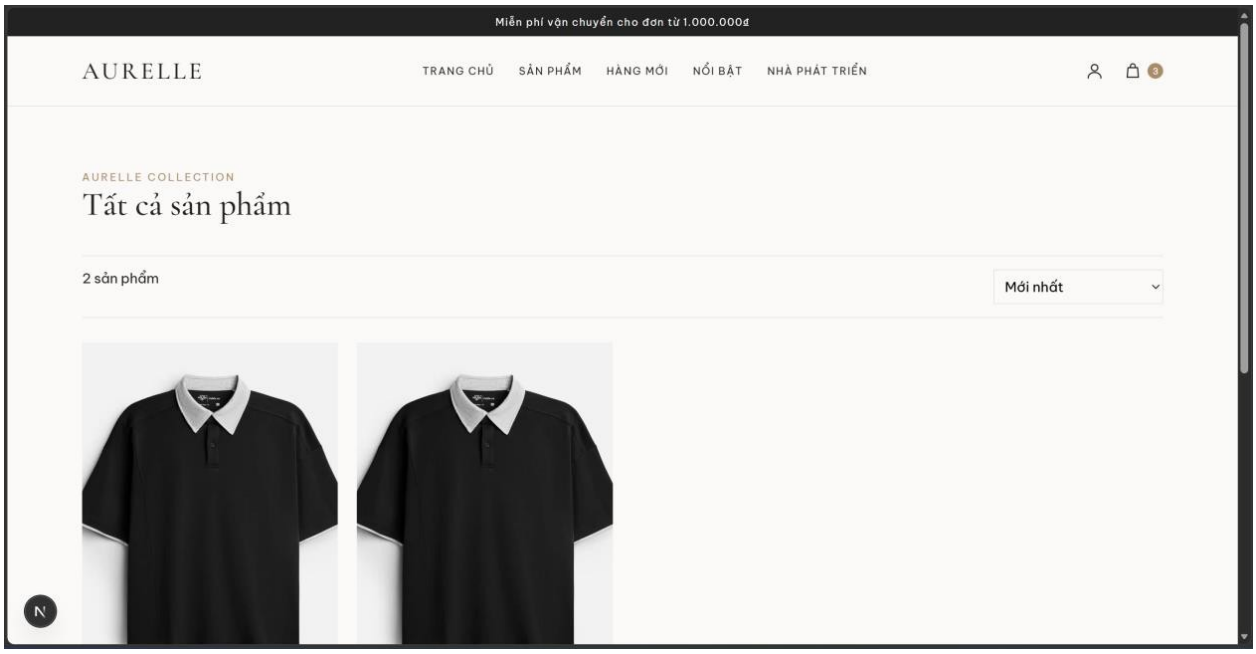
3.11.1 Browse, cart and checkout

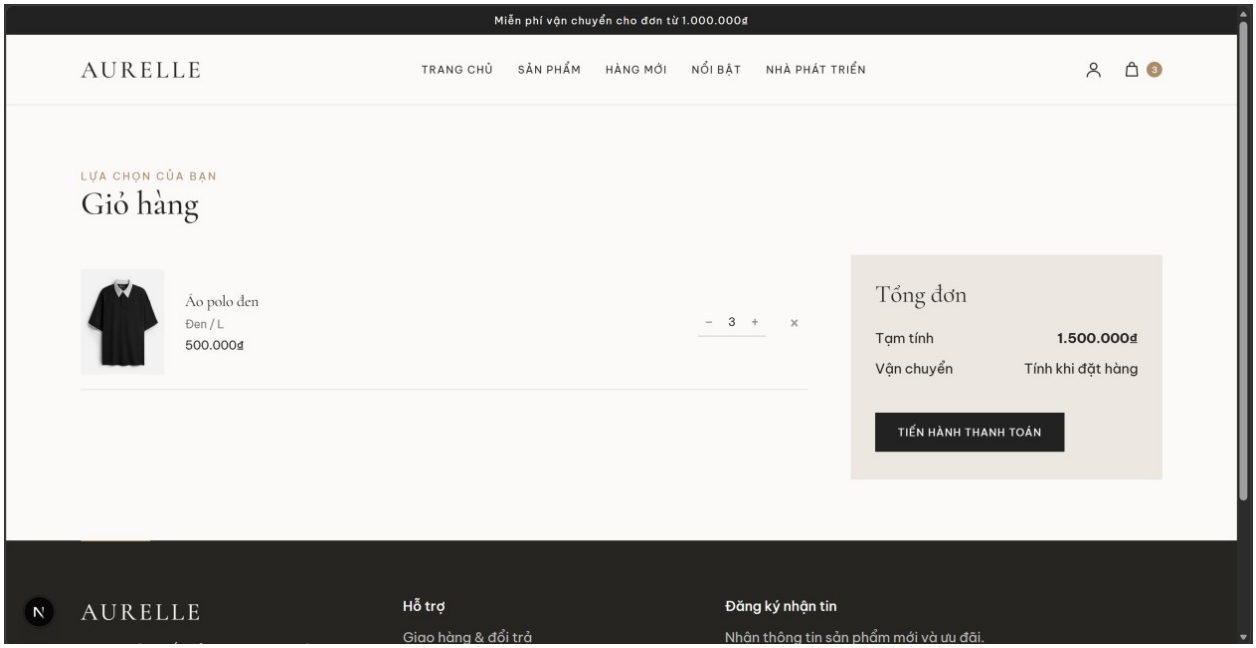
Function trigger: Home → Shop → Product → Cart → Checkout.

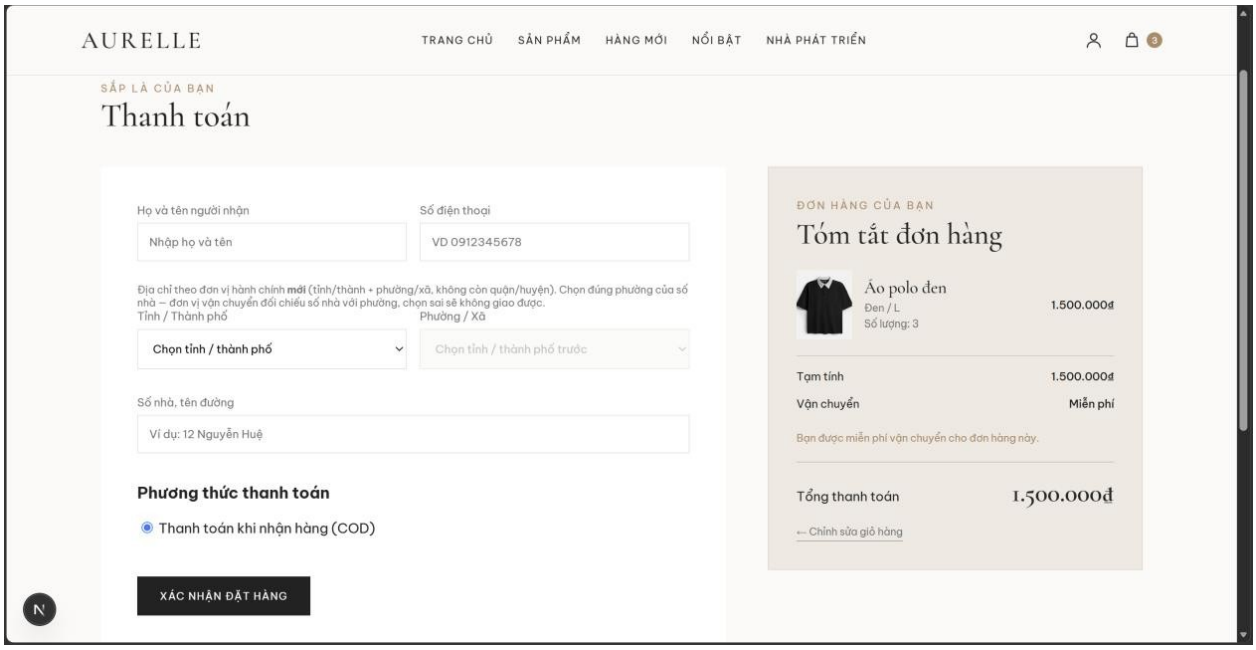
Function description: Customer.

Screen layout:


Miễn phí vận chuyển cho đơn từ 1.000.000đ
AURELLE
TRANG CHỦ
SẢN PHẨM
HÀNG MỚI
NỔI BẬT
NHÀ PHÁT TRIỂN
SPRING / SUMMER 2026
Essentials for every day.
Những thiết kế tối giản, chất liệu dễ chịu và đủ linh hoạt để đồng hành cùng bạn mỗi ngày.
KHÁM PHÁ BỘ SƯU TẬP


Miễn phí vận chuyển cho đơn từ 1.000.000đ
AURELLE
TRANG CHỦ
SẢN PHẨM
HÀNG MỚI
NỔI BẬT
NHÀ PHÁT TRIỂN
AURELLE COLLECTION
Tất cả sản phẩm
2 sản phẩm
Mới nhất





Function details:

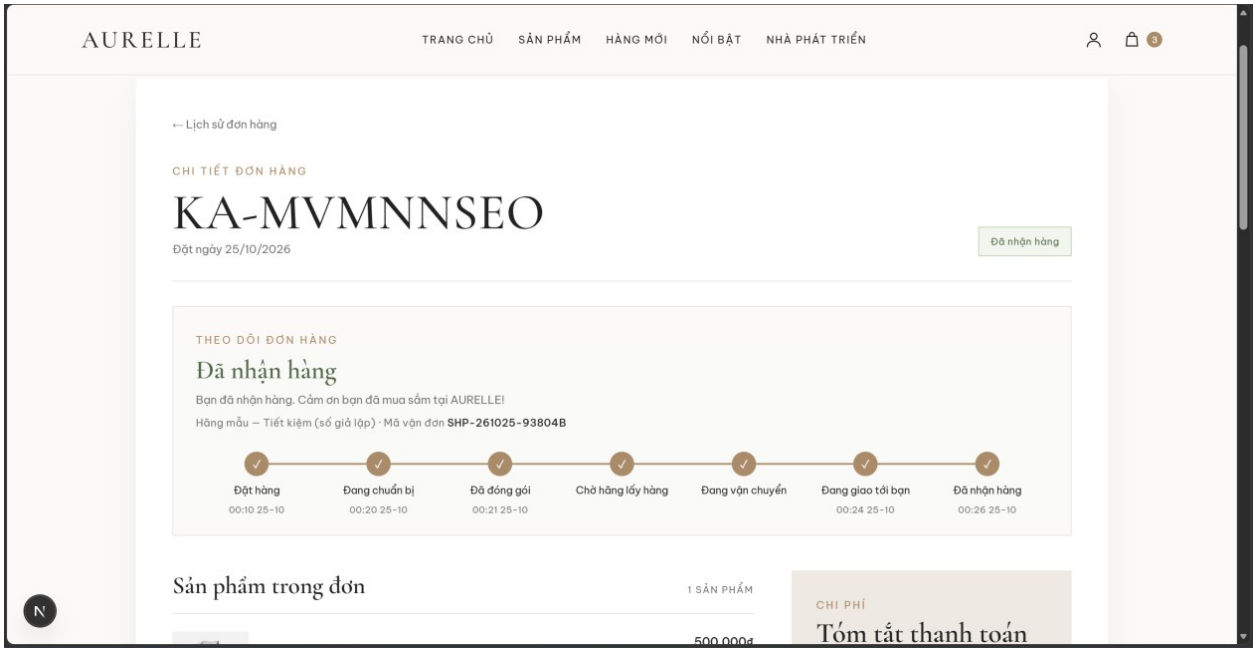
- Only variants whose stock is managed by the OptiPack warehouse are sellable.
- Address uses the new two-level administrative units (province, ward); phone must be 10 digits starting with 0.
- Payment method: COD or bank transfer.

3.11.2 Track and cancel order

Function trigger: My Orders → Order Detail.

Function description: Customer.

Screen layout:



Function details:

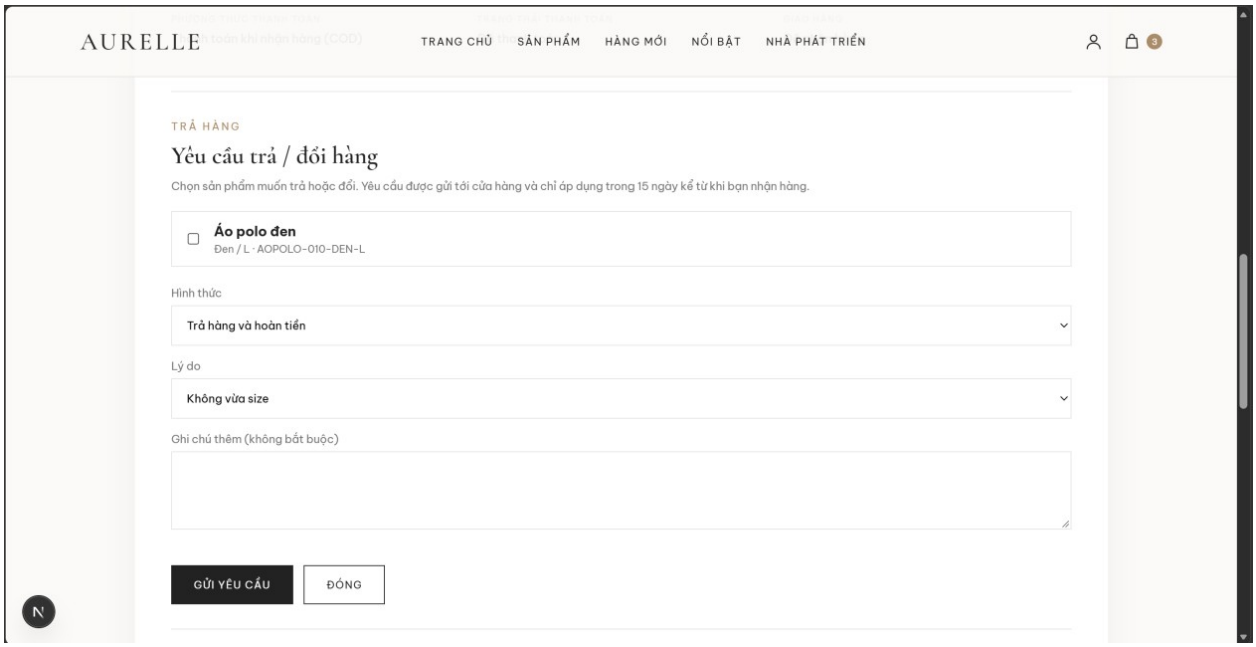
- A 7-step progress bar (Placed, Preparing, Packed, Waiting for pickup, In transit, Out for delivery, Delivered) is shown from the moment the order is placed, with branches for cancellation, failed delivery and return.
- The page refreshes every 30 seconds while the order is not finished.
- The customer can cancel only while the order is pending and not packed.

3.11.3 Return and exchange request

Function trigger: Order Detail → Request return.

Function description: Customer.

Screen layout:



Function details:

- Allowed within 15 days after delivery, one open request per order; exchange target must be in stock.
- A pending request can be cancelled by the customer.

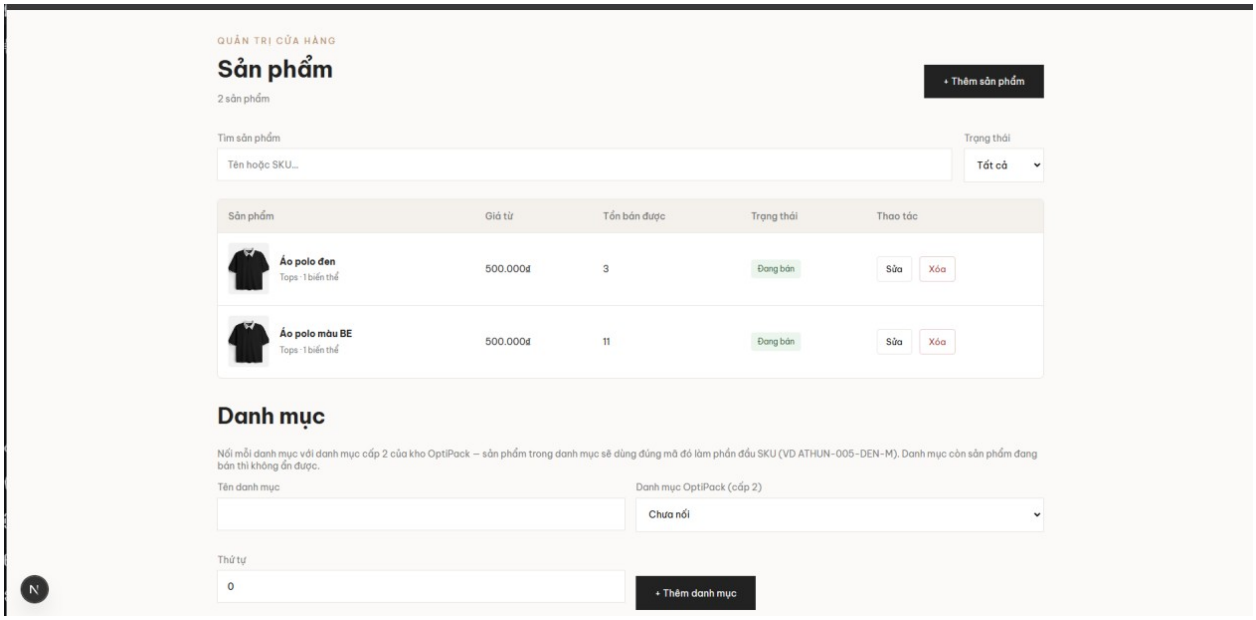
3.12 AURELLE Web Administration

3.12.1 Manage storefront catalog

Function trigger: AURELLE Web Admin page.

Function description: AURELLE Web Admin.

Screen layout:



Function details:

- New variants must use OptiPack category, model number, colour and size so the SKU matches an internal SKU.
- Saving a product notifies OptiPack (product-updated webhook) so the catalog is synchronized.

4. Non-Functional Requirements

4.1 External Interfaces

Interface	Description
User interfaces	Web portal for staff (web browser; a mobile scanning app is planned for Iteration 3 and is not part of this release), AURELLE customer website and admin page (desktop and mobile browser).
Lazada Open Platform	HTTPS REST with OAuth 2.0 access / refresh tokens and request signing.
AURELLE Open API	HTTPS REST in Lazada style with HMAC-SHA256 signed requests; webhooks signed with the app secret. In the demo the Open API runs as a simulator (be/scripts/aurelle-mock-server.ts, port 4000) that reads the AURELLE website data.
GHN API	HTTPS REST with token and shop id; webhooks for status changes; A5 shipping labels.
Google OAuth 2.0	Login for existing staff accounts.
SMTP (Gmail)	Transactional e-mails.
CP-SAT packer service	Internal HTTP service (Python, OR-Tools) used to prove packing plans; optional.
AI text service	Groq API (OpenAI-compatible chat completions, model openai/gpt-oss-120b, timeout 20 s) used to word packing guides; the template text is used when it is unavailable. No customer data is sent.

4.2 Quality Attributes

4.2.1 Usability

- Each role lands on its own home screen and sees only its menu items.
- Every error returned to the user has a stable error code and a readable Vietnamese message.
- Picking and packing screens on the web portal accept barcode scanner input and manual entry.

4.2.2 Reliability

Requirement	Metric / threshold	How it is achieved / measured
Stock never goes negative or double-counted	0 negative bin stock	Atomic conditional decrement in a MongoDB transaction; idempotent scans by client event id; unit tests.
No duplicated external events	0 duplicates	Processed webhook / carrier event collections and idempotency keys on write-back.
Status write-back to AURELLE	Retried up to 8 times with back-off 30 s → 30 min	Outbox written in the same transaction as the business change.
Degraded mode when packer service is down	Plans still produced	Fallback to the built-in BRKGA solver (label "heuristic").
Automated regression	≥ 1,251 automated back-end tests pass on every push	GitHub Actions CI: type check, lint, tests, builds.

4.2.3 Performance

Requirement	Metric / threshold	How it is achieved / measured
New Lazada orders appear in OptiPack	≤ configured sync interval + 1 minute (default ≤ 6 minutes)	Scheduler every minute; each shop is synchronized when its interval (default 5 minutes) has passed.
Picked group gets a packing plan	Job runs every 10 s; BRKGA p95 ≤ 1 s per order; CP-SAT limit 6 s per order	Benchmark scripts/solver-benchmark.ts on 300 orders (last run p95 560 ms).
Customer sees status change on AURELLE	≤ 15 s write-back + ≤ 30 s page refresh	Write-back job every 15 s; page auto-refresh.
AURELLE stock follows warehouse stock	≤ 1 minute	Stock push every minute.
API protection	600 requests / minute / client	Global rate limiter.

4.2.4 Security

- Passwords hashed with bcrypt; account lockout after 5 failed logins for 15 minutes.
- JWT access token 24 hours, refresh token 30 days with rotation and reuse detection; logout revokes the refresh token.
- Optional two-factor authentication (TOTP + 10 one-time backup codes) for every staff account; trusted device 30 days.
- Marketplace tokens encrypted with AES-256-GCM; AURELLE and GHN callbacks verified by signature / secret.
- Role-based access control on every endpoint; HTTP security headers (Helmet); login audit log.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Orders of the same shop with the same consolidation key are grouped while the group is still

ID	Rule Definition
	awaiting_packaging or picking (configurable).
BR-02	Each parcel contains items of exactly one order; each parcel has its own carrier tracking code with the recipient of that order.
BR-03	Split or merge of order groups is allowed only before the first item is scanned; merge requires same shop and same real phone number.
BR-04	Stock is reserved per order group; picking consumes the oldest lot first (FIFO).
BR-05	A group can be marked picked only when all ordered quantities are scanned; otherwise missing items must be reported.
BR-06	A packing plan cannot be approved while any item is unplaced; a manual plan must be approved by another person.
BR-07	A parcel whose weight differs from the estimate by more than the tolerance (default 20%) is held and reviewed by a different person.
BR-08	Fragile items are packed in new boxes unless reused boxes are allowed in packing settings.
BR-09	A rejected packing plan must be handled within 2 working hours (08:00–17:00).
BR-10	Without carrier API, a shipment returns to the warehouse after 2 failed delivery attempts or customer refusal.
BR-11	Return requests are accepted within 15 days after delivery; one open request per order.
BR-12	The creator of a return ticket cannot approve it; reject and cancel require a reason.
BR-13	A customer can cancel an AURELLE order only while it is pending and not packed; the seller can cancel until it is handed to the carrier.
BR-14	Short-shipped items are allocated to the oldest orders first and are not charged.
BR-15	Only the six clothing categories can be folded in half in a packing plan.
BR-16	Staff on leave are not assigned work automatically or manually.
BR-17	A new account must change its temporary password within 72 hours.

5.2 Common Requirements

- Every state-changing request that can be performed by two people at the same time carries the version of the record (expected_version); an outdated version is rejected with a conflict error.
- Errors are returned as { error_code, message, details } with HTTP status 400 / 401 / 403 / 404 / 409 / 422.
- All dates are stored in UTC and displayed in Asia/Ho_Chi_Minh time.
- Every business action that changes the state of an order group, return ticket or stock is recorded in the history.

5.3 Application Messages List

Code	Type	Context	Content
AUTH_INVALID_CREDENTIALS	Error	Login	Email hoặc mật khẩu không chính xác. (Wrong e-mail or password.)
AUTH_ACCOUNT_LOCKED	Error	Login	Account locked for 15 minutes after too many failed attempts.
AUTH_PASSWORD_DEADLINE_LOCKED	Error	Login	Account locked: temporary password not changed within 72 hours.
ORD_GROUP_STATE_CONFLICT	Error	Any group action	The group was changed by someone else — reload and try again.
ORD_GROUP_PICK_EXCEEDS_ORDERED	Error	Picking	Scanned quantity exceeds the ordered quantity.
ORD_GROUP_INSUFFICIENT_STOCK	Error	Picking	Not enough stock in the bin — report missing items.
ORD_GROUP_PICK_INCOMPLETE	Error	Picking	Some SKUs are not fully picked.
ORD_GROUP_WRONG_WAR	Error	Picking	This round is being picked in another warehouse.

Code	Type	Context	Content
EHOUSE			
ORD_GROUP_REGROUP_HAS_PICKS	Error	Split / merge	Items already scanned — the group cannot be split or merged.
ORD_GROUP_MERGE_RECIPIENT_MISMATCH	Error	Merge	Different phone numbers — use "Link" instead of merging.
PACKING_SCAN_REQUIRED	Error	Quick pack	Scanning each item is required (setting or several recipients).
PACKING_PACK_WEIGHTS_MISMATCH	Error	Packing	Weights must be entered once for every unsealed parcel.
PKG_BOX_OUT_OF_STOCK	Error	Packing	The selected box is out of stock.
WH_SKU_IN_OTHER_BIN	Confirmation	Bin assignment	The SKU is already in another bin — confirm to use a second bin.
RMA_WINDOW_EXPIRED	Error	Returns	More than 15 days since delivery — return period expired.
RMA_SIMULATION_DISABLED	Error	Returns	Simulated returns are disabled; customers request returns on AURELLE.
PM_FOLD_NOT_ALLOWED	Error	Packaging profile	Only clothing categories can be folded in half.
consolidation_name_mismatch	Notification	Order groups	Consolidated group has several recipients with different names.
stock_shortage	Notification	Order groups	Available stock does not cover the group.
missing_item	Notification	Picking	Warehouse Staff reported a missing item.
packing_blocked	Notification	Packing	Packing plan blocked — SKU needs measurement.
delivery_failed	Notification	Shipping	Carrier reported a failed delivery.
return_requested	Notification	Returns	A customer requested a return / exchange.

5.4 Other Requirements

The main business workflows WF-01 to WF-06 that combine the features above (actors, triggers, outcomes and state changes) are specified with activity diagrams in Report 4 – Software Design Document.